

Untitled

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```
library(Lahman)
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.1.2
```

```
## Warning: package 'tibble' was built under R version 4.1.2
```

```
## Warning: package 'forcats' was built under R version 4.1.2
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
library(car)
```

```
## Warning: package 'car' was built under R version 4.1.2
```

```
## Loading required package: carData
```

```
## Warning: package 'carData' was built under R version 4.1.2
```

```
##
```

```
## Attaching package: 'carData'
```

```
##
```

```
## The following object is masked from 'package:Lahman':
```

```
##
```

```
##      Salaries
```

```
##
```

```
##
```

```
## Attaching package: 'car'
```

```
##
```

```
## The following object is masked from 'package:dplyr':
##
##   recode
##
## The following object is masked from 'package:purrr':
##
##   some
```

```
data(Pitching)
```

```
head(Pitching)
```

```
##   playerID yearID stint teamID lgID W L  G GS CG SHO SV IPouts  H ER HR BB SO
## 1 aardsda01  2004     1   SFN  NL 1 0 11  0  0  0  0    32 20  8  1 10  5
## 2 aardsda01  2006     1   CHN  NL 3 0 45  0  0  0  0   159 41 24  9 28 49
## 3 aardsda01  2007     1   CHA  AL 2 1 25  0  0  0  0    97 39 23  4 17 36
## 4 aardsda01  2008     1   BOS  AL 4 2 47  0  0  0  0   146 49 30  4 35 49
## 5 aardsda01  2009     1   SEA  AL 3 6 73  0  0  0 38   214 49 20  4 34 80
## 6 aardsda01  2010     1   SEA  AL 0 6 53  0  0  0 31   149 33 19  5 25 49
##   BAOpp ERA  IBB WP  HBP BK  BFP GF  R SH SF GIDP
## 1 0.417 6.75   0  0   2  0  61  5  8  0  1    1
## 2 0.214 4.08   0  1   1  0 225  9 25  1  3    2
## 3 0.300 6.40   3  2   1  0 151  7 24  2  1    1
## 4 0.268 5.55   2  3   5  0 228  7 32  3  2    4
## 5 0.190 2.52   3  2   0  0 296 53 23  2  1    2
## 6 0.198 3.44   5  2   2  0 202 43 19  7  1    5
```

```
data(pitchingLabels)
```

```
pitchingLabels
```

```
##   variable                                label
## 1  playerID                            Player ID code
## 2   yearID                               Year
## 3   stint                               Player's stint
## 4   teamID                               Team
## 5    lgID                               League
## 6     W                                  Wins
## 7     L                                  Losses
## 8     G                                  Games
## 9     GS                                Games Started
## 10    CG                                Complete Games
## 11    SHO                               Shutouts
## 12    SV                               Saves
## 13   IPouts                             Outs Pitched
## 14    H                                  Hits
## 15    ER                                Earned Runs
## 16    HR                                Homeruns
## 17    BB                                Walks
## 18    SO                                Strikeouts
## 19   BAOpp                            Opponent's Batting Average
## 20    ERA                               Earned Run Average
## 21    IBB                               Intentional Walks
```

```
## 22      WP                               Wild Pitches
## 23      HBP                            Batters Hit By Pitch
## 24      BK                               Balks
## 25      BFP                            Batters faced by Pitcher
## 26      GF                               Games Finished
## 27      R                               Runs Allowed
## 28      SH                            Sacrifices by opposing batters
## 29      SF                            Sacrifice flies by opposing batters
## 30      GIDP Grounded into double plays by opposing batter
```

```
Pitching_full <- Pitching

Pitching_full <- Pitching_full %>%
  filter(GS > 0, IPouts > 100, yearID > 1999) %>%
  mutate(SOrate = SO / IPouts, HRper9 = HR * 27 / IPouts, OutRate = BFP / IPouts)

head(Pitching_full)
```

```
##   playerID yearID stint teamID lgID  W L  G GS CG SHO SV IPouts  H ER HR BB
## 1 abadfe01  2012     1   HOU  NL   0 6 37  6  0  0  0   138  57 26  6 19
## 2 abbotan01  2023     1   CIN  NL   8 6 21 21  0  0  0   328 100 47 16 44
## 3 abbotco01  2022     1   WAS  NL   0 5 16  9  0  0  0   144  44 28 12 25
## 4 abbotpa01  2000     1   SEA  AL   9 7 35 27  0  0  0   537 164 84 23 80
## 5 abbotpa01  2001     1   SEA  AL  17 4 28 27  1  0  0   489 145 77 21 87
## 6 abbotpa01  2003     1   KCA  AL   1 2 10  8  0  0  0   143  47 28  8 26
##   SO BAOpp  ERA IBB WP HBP BK BFP GF  R SH SF GIDP   SOrate  HRper9  OutRate
## 1  38 0.311 5.09  1  4  3  0 208  8 27  2  1    4 0.2753623 1.173913 1.507246
## 2 120 0.244 3.87  0  1  1  1 459  0 47  0  5    6 0.3658537 1.317073 1.399390
## 3  45 0.239 5.25  0  2  5  0 216  7 30  0  2    0 0.3125000 2.250000 1.500000
## 4 100 0.243 4.22  4  3  5  0 766  2 89  1  4   20 0.1862197 1.156425 1.426443
## 5 118 0.238 4.25  5 11  7  0 710  0 79  3  5   11 0.2413088 1.159509 1.451943
## 6  32 0.257 5.29  2  2  2  0 214  0 29  2  1    1 0.2237762 1.510490 1.496503
```

```
Pitching_data <- na.omit(Pitching_full)
Pitching_data <- Pitching_data[-c(1, 2, 3, 4, 5)]

head(Pitching_data)
```

```
##   W L  G GS CG SHO SV IPouts  H ER HR BB  SO BAOpp  ERA IBB WP HBP BK BFP GF
## 1  0 6 37  6  0  0  0   138  57 26  6 19  38 0.311 5.09  1  4  3  0 208  8
## 2  8 6 21 21  0  0  0   328 100 47 16 44 120 0.244 3.87  0  1  1  1 459  0
## 3  0 5 16  9  0  0  0   144  44 28 12 25  45 0.239 5.25  0  2  5  0 216  7
## 4  9 7 35 27  0  0  0   537 164 84 23 80 100 0.243 4.22  4  3  5  0 766  2
## 5 17 4 28 27  1  0  0   489 145 77 21 87 118 0.238 4.25  5 11  7  0 710  0
## 6  1 2 10  8  0  0  0   143  47 28  8 26  32 0.257 5.29  2  2  2  0 214  0
##   R SH SF GIDP   SOrate  HRper9  OutRate
## 1 27  2  1    4 0.2753623 1.173913 1.507246
## 2 47  0  5    6 0.3658537 1.317073 1.399390
## 3 30  0  2    0 0.3125000 2.250000 1.500000
## 4 89  1  4   20 0.1862197 1.156425 1.426443
## 5 79  3  5   11 0.2413088 1.159509 1.451943
## 6 29  2  1    1 0.2237762 1.510490 1.496503
```

```
library(Boruta)
```

```
## Warning: package 'Boruta' was built under R version 4.1.2
```

```
response_variable <- "W"
```

```
predictors <- setdiff(names(Pitching_data), response_variable)
```

```
boruta_formula <- as.formula(paste(response_variable, "~ ."))
```

```
boruta_result <- Boruta(boruta_formula, data = Pitching_data, doTrace = 2)
```

```
## 1. run of importance source...
```

```
## 2. run of importance source...
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```
## 3. run of importance source...
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```
## 4. run of importance source...
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```
## 5. run of importance source...
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## 6. run of importance source...
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## 7. run of importance source...
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```
## 8. run of importance source...
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## 9. run of importance source...
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## 10. run of importance source...
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```
## 11. run of importance source...
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```
## 12. run of importance source...
```

```
## After 12 iterations, +37 secs:
```

```
## confirmed 25 attributes: BAOpp, BB, BFP, CG, ER and 20 more;
```

```
## rejected 1 attribute: BK;
```

```
## still have 1 attribute left.
```

```
## 13. run of importance source...
```

```
## 14. run of importance source...
```

15. run of importance source...

16. run of importance source...

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```
## 87. run of importance source...

## 88. run of importance source...

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## 90. run of importance source...

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## 95. run of importance source...

## 96. run of importance source...

## 97. run of importance source...

## 98. run of importance source...

## 99. run of importance source...
```

```
boruta_result
```

```
## Boruta performed 99 iterations in 4.830891 mins.
## 25 attributes confirmed important: BAOpp, BB, BFP, CG, ER and 20 more;
## 1 attributes confirmed unimportant: BK;
## 1 tentative attributes left: SV;
```

```
attribute_stats <- attStats(boruta_result)
top_5_attributes <- attribute_stats[order(-attribute_stats$meanImp), ]
top_5_attributes
```

##	meanImp	medianImp	minImp	maxImp	normHits	decision
## L	38.7198838	38.4151874	35.2657234	44.066544	1.0000000	Confirmed
## ERA	28.1895663	28.1003870	25.3616444	31.705978	1.0000000	Confirmed
## OutRate	26.4036020	26.3225021	23.1844139	29.419374	1.0000000	Confirmed
## BAOpp	21.9154748	21.8817558	19.3157528	24.438373	1.0000000	Confirmed
## IPouts	20.5856450	20.5881243	19.5501463	21.930626	1.0000000	Confirmed
## BFP	18.4736368	18.4830244	16.6087976	20.140674	1.0000000	Confirmed
## GS	15.8165145	15.8860616	14.0780798	17.485221	1.0000000	Confirmed
## SO	15.1664405	15.2198873	13.0703715	16.872195	1.0000000	Confirmed
## H	14.1722536	14.1879795	12.0828609	16.027049	1.0000000	Confirmed
## HRper9	14.1301580	14.1784487	10.0184326	16.911434	1.0000000	Confirmed
## GIDP	13.7422037	13.6764734	10.4001810	16.514777	1.0000000	Confirmed

## R	13.5474569	13.6374659	6.8759513	15.536761	1.0000000	Confirmed
## SOrate	13.3326253	13.3842613	10.7332728	16.049572	1.0000000	Confirmed
## BB	13.1898220	13.4717686	7.0986970	15.102308	1.0000000	Confirmed
## ER	12.5778393	12.6255722	8.9856443	14.344737	1.0000000	Confirmed
## G	11.6871858	11.7890061	8.9420947	14.097685	1.0000000	Confirmed
## HR	9.7921644	9.8664077	7.2748993	12.449183	1.0000000	Confirmed
## CG	9.4636647	9.3164614	6.9434081	12.238684	1.0000000	Confirmed
## SH	9.1839044	9.1179236	7.0769581	11.674194	1.0000000	Confirmed
## HBP	8.7371950	8.7595950	6.6398943	11.302835	1.0000000	Confirmed
## SF	8.4605812	8.4190894	5.5955784	11.027984	1.0000000	Confirmed
## GF	7.7695771	7.7532111	4.4433560	11.903834	1.0000000	Confirmed
## SHO	6.5958258	6.6805725	4.2540680	8.518617	1.0000000	Confirmed
## WP	5.6081843	5.6577572	3.8997054	8.043000	1.0000000	Confirmed
## IBB	5.4873638	5.5086467	2.6840907	7.959996	0.9797980	Confirmed
## SV	2.2049611	2.2727864	0.4786063	4.128081	0.5050505	Tentative
## BK	-0.3499129	-0.2719718	-1.8215714	1.357674	0.0000000	Rejected

Upon Running the Boruta Test, We find that 26 of our attributes are considered important, and that only one (Balks) do not contribute to wins. Upon looking at the attributes ordered by mean importance, we have decided to use L, Outrate, ERA, BAOpp, and SO. We did not use IPouts because Outrate is based on IPouts so we will run into a multicollinearity problem, and we did not use BFP due to its similarity with IPouts. We also rejected GS because $W + L$ are almost always equal to GS, and that would lead to problems later on.

```
summary(lm(W ~ teamID, data = Pitching_full)) #R^2 0.02
```

```
##
## Call:
## lm(formula = W ~ teamID, data = Pitching_full)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8.197 -3.829 -1.005   3.385 17.746
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   7.48837    0.71957  10.407 < 2e-16 ***
## teamIDARI     -0.58631    0.79533  -0.737  0.46103
## teamIDATL      0.68112    0.80223   0.849  0.39590
## teamIDBAL     -1.65910    0.79145  -2.096  0.03610 *
## teamIDBOS      0.22017    0.79351   0.277  0.78143
## teamIDCHA      0.07274    0.80092   0.091  0.92764
## teamIDCHN     -0.06530    0.80007  -0.082  0.93496
## teamIDCIN     -1.51362    0.79387  -1.907  0.05662 .
## teamIDCLE     -0.30376    0.79496  -0.382  0.70240
## teamIDCOL     -1.39792    0.79351  -1.762  0.07817 .
## teamIDDET     -1.23471    0.79145  -1.560  0.11880
## teamIDFLO     -0.15858    0.86870  -0.183  0.85515
## teamIDHOU      0.04073    0.79723   0.051  0.95926
## teamIDKCA     -1.92378    0.79013  -2.435  0.01493 *
## teamIDLAA     -0.40142    0.80998  -0.496  0.62020
## teamIDLAN      0.12549    0.79246   0.158  0.87418
## teamIDMIA     -2.68837    0.85430  -3.147  0.00166 **
```

```
## teamIDMIL -0.61203 0.79842 -0.767 0.44338
## teamIDMIN -0.48302 0.79802 -0.605 0.54502
## teamIDMON -1.53383 1.01182 -1.516 0.12960
## teamIDNYA 0.46640 0.79351 0.588 0.55671
## teamIDNYN -0.39795 0.79763 -0.499 0.61786
## teamIDOAK -0.39043 0.79533 -0.491 0.62351
## teamIDPHI -0.48316 0.79608 -0.607 0.54392
## teamIDPIT -1.76560 0.79246 -2.228 0.02592 *
## teamIDSDN -1.42305 0.79351 -1.793 0.07297 .
## teamIDSEA -0.48316 0.79608 -0.607 0.54392
## teamIDSFN 0.12615 0.80135 0.157 0.87491
## teamIDSLN 0.70844 0.79763 0.888 0.37448
## teamIDTBA -1.12122 0.79078 -1.418 0.15628
## teamIDTEX -1.27510 0.78949 -1.615 0.10635
## teamIDTOR -0.28202 0.79723 -0.354 0.72354
## teamIDWAS -1.09764 0.81561 -1.346 0.17842
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.719 on 5812 degrees of freedom
## Multiple R-squared: 0.02669, Adjusted R-squared: 0.02133
## F-statistic: 4.98 on 32 and 5812 DF, p-value: < 2.2e-16
```

```
summary(lm(W ~ lgID, data = Pitching_full)) #R^2 -0.0001
```

```
##
## Call:
## lm(formula = W ~ lgID, data = Pitching_full)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.8736 -3.8617 -0.8736  3.1383 17.1383
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  6.87365    0.08912   77.130 <2e-16 ***
## lgIDNL      -0.01190    0.12481  -0.095  0.924
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.77 on 5843 degrees of freedom
## Multiple R-squared: 1.556e-06, Adjusted R-squared: -0.0001696
## F-statistic: 0.009095 on 1 and 5843 DF, p-value: 0.924
```

```
summary(lm(W ~ yearID, data = Pitching_full))
```

```
##
## Call:
## lm(formula = W ~ yearID, data = Pitching_full)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.918 -3.828 -1.024  3.255 17.074
```

```
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 188.372280  17.672363   10.66  <2e-16 ***
## yearID      -0.090227   0.008785  -10.27  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.728 on 5843 degrees of freedom
## Multiple R-squared:  0.01773,    Adjusted R-squared:  0.01757
## F-statistic: 105.5 on 1 and 5843 DF,  p-value: < 2.2e-16
```

Upon looking at our Datasets three factor variables, we have decided to use lgID and teamID. Neither have strong R^2 values, but the third factor, yearID, may work better as a discrete variable due to its counting nature.

Part 2

```
fivenum(Pitching_data$L)
```

```
## [1]  0  4  6  9 21
```

```
fivenum(Pitching_data$ERA)
```

```
## [1]  0.95  3.62  4.36  5.20 10.69
```

```
fivenum(Pitching_data$BAOpp)
```

```
## [1] 0.129 0.241 0.263 0.284 0.395
```

```
fivenum(Pitching_data$SO)
```

```
## [1] 10  48  80 128 372
```

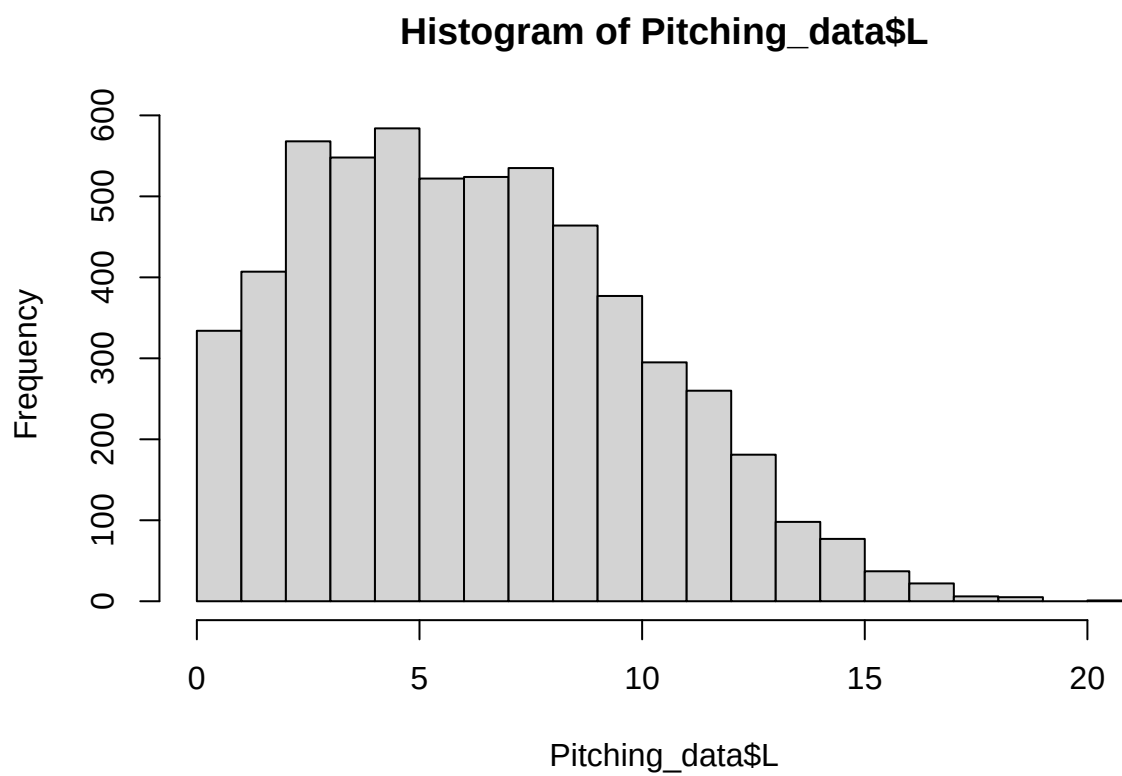
```
fivenum(Pitching_data$OutRate)
```

```
## [1] 1.173913 1.388554 1.430108 1.481050 1.792453
```

```
fivenum(Pitching_data$W)
```

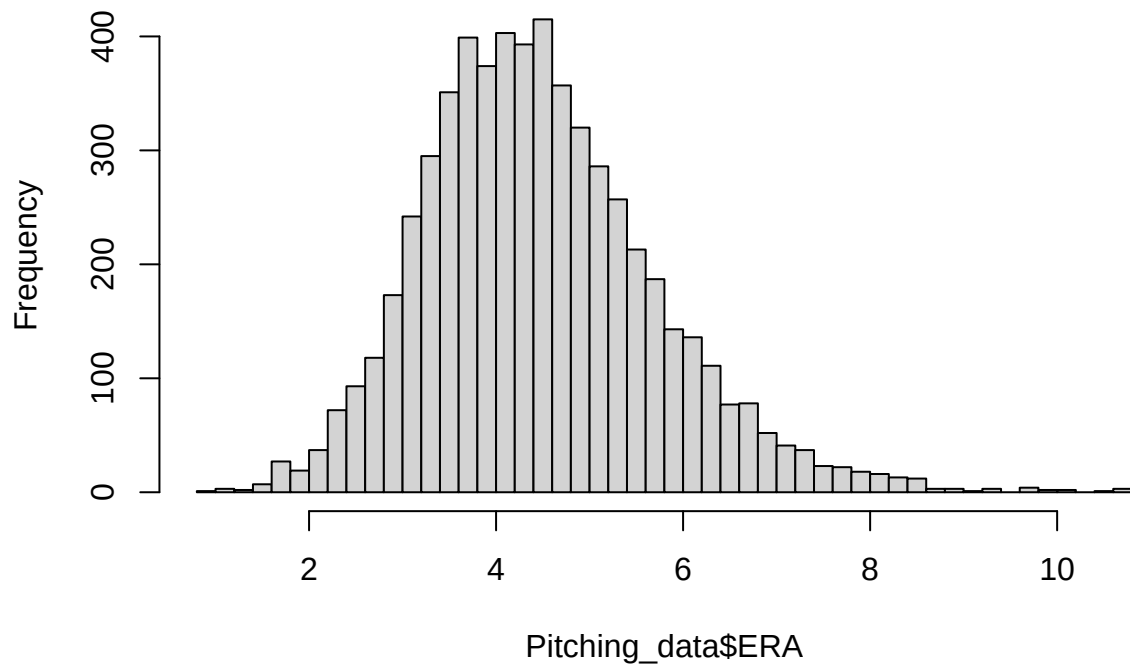
```
## [1]  0  3  6 10 24
```

```
hist(Pitching_data$L, breaks = 18)
```



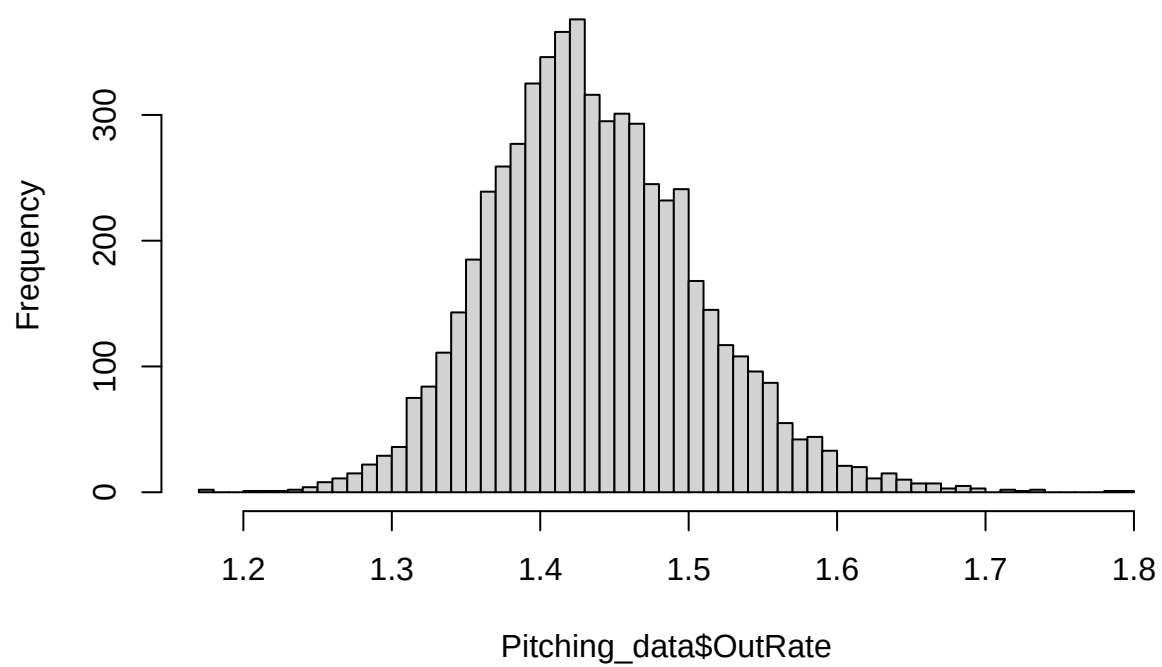
```
hist(Pitching_data$ERA, breaks = "scott")
```

Histogram of Pitching_data\$ERA



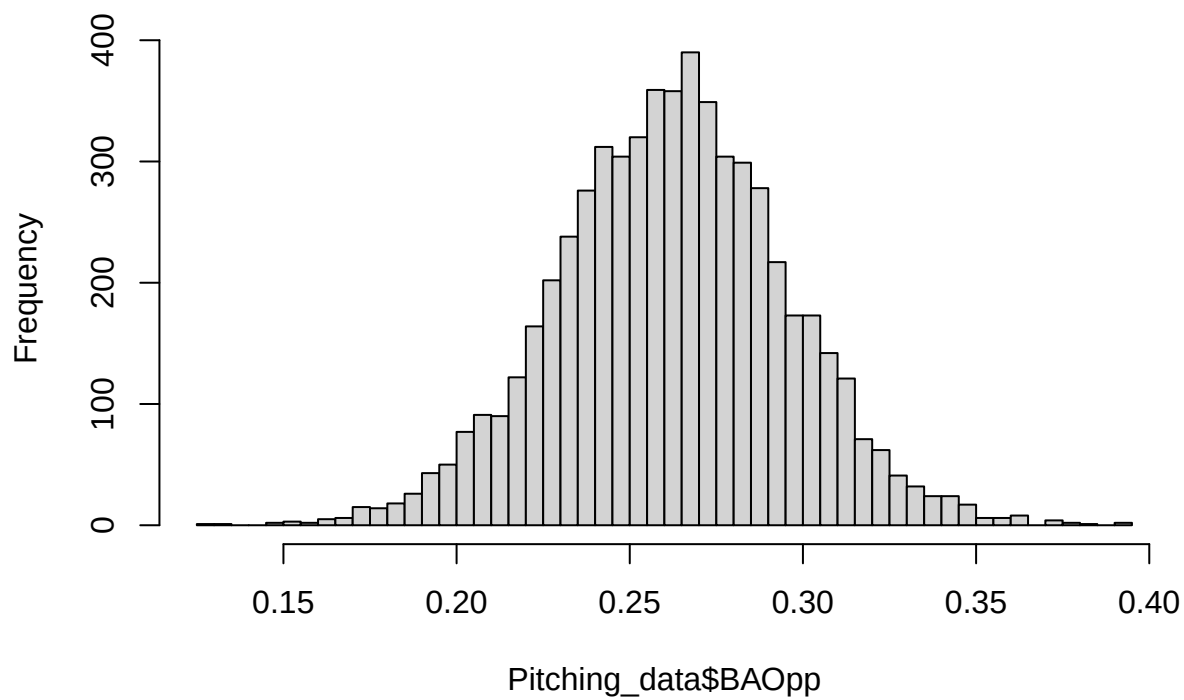
```
hist(Pitching_data$OutRate, breaks = "scott")
```

Histogram of Pitching_data\$OutRate



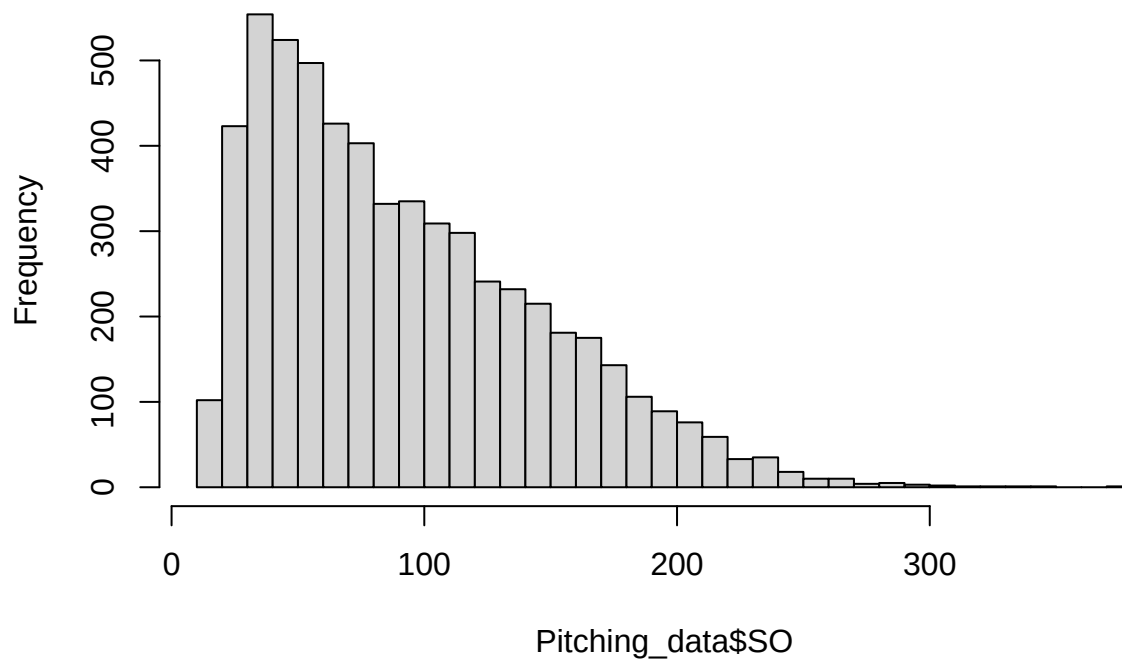
```
hist(Pitching_data$BAOpp, breaks = "scott")
```

Histogram of Pitching_data\$BAOpp



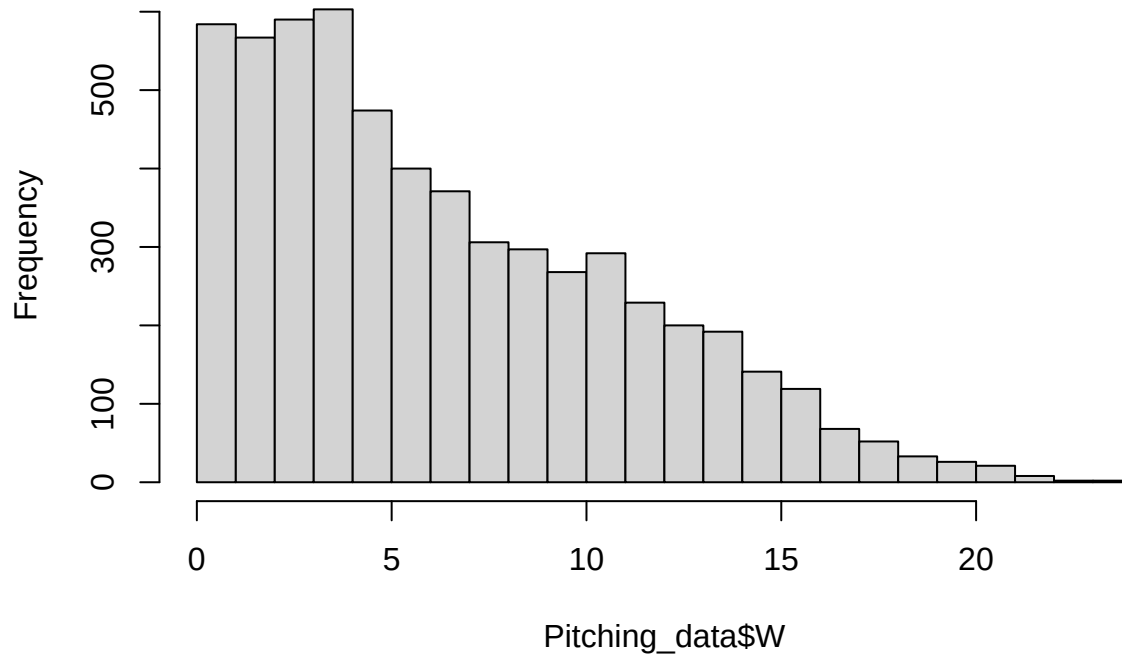
```
hist(Pitching_data$S0, breaks = "scott")
```

Histogram of Pitching_data\$SO

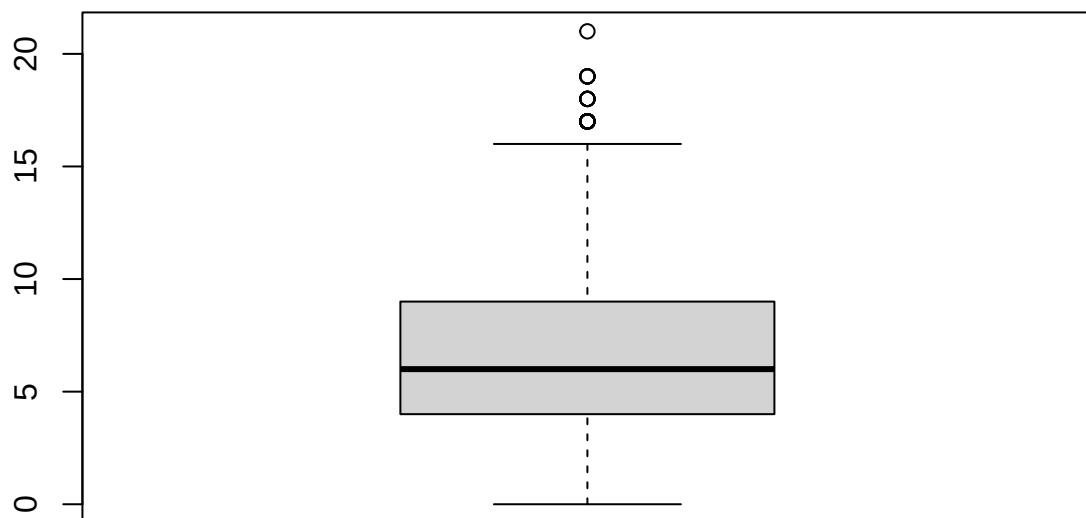


```
hist(Pitching_data$W, breaks = 18)
```

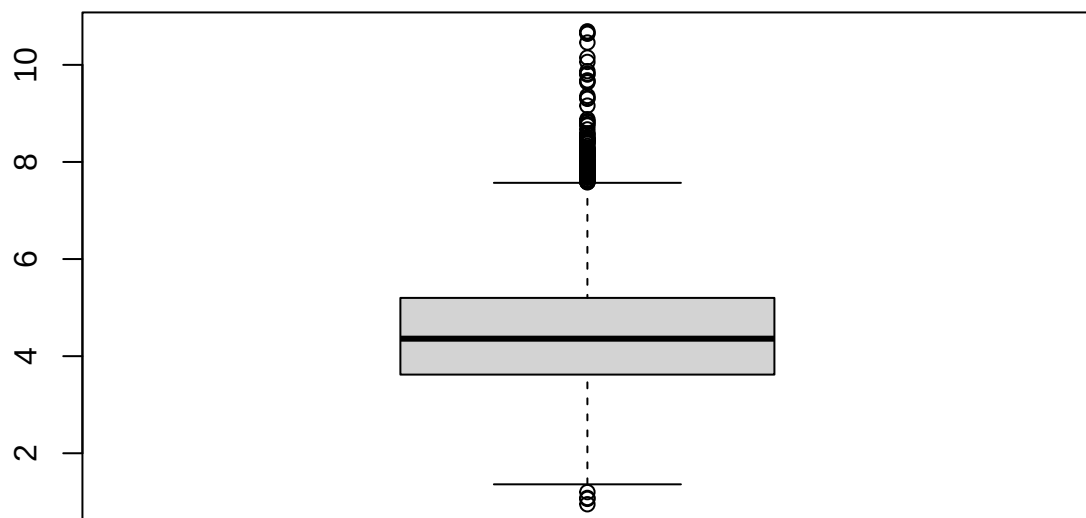

Histogram of Pitching_data\$W



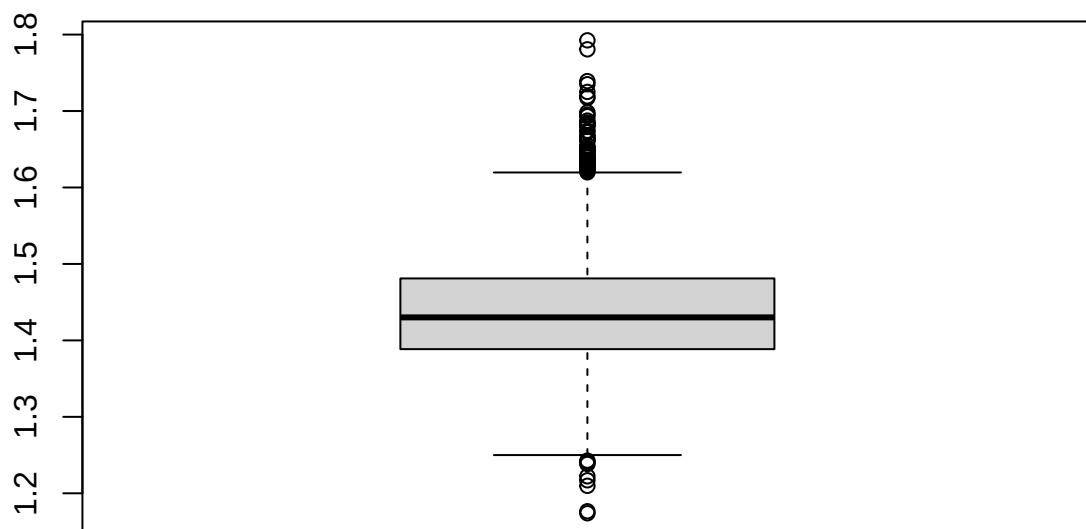
```
boxplot(Pitching_data$L)
```



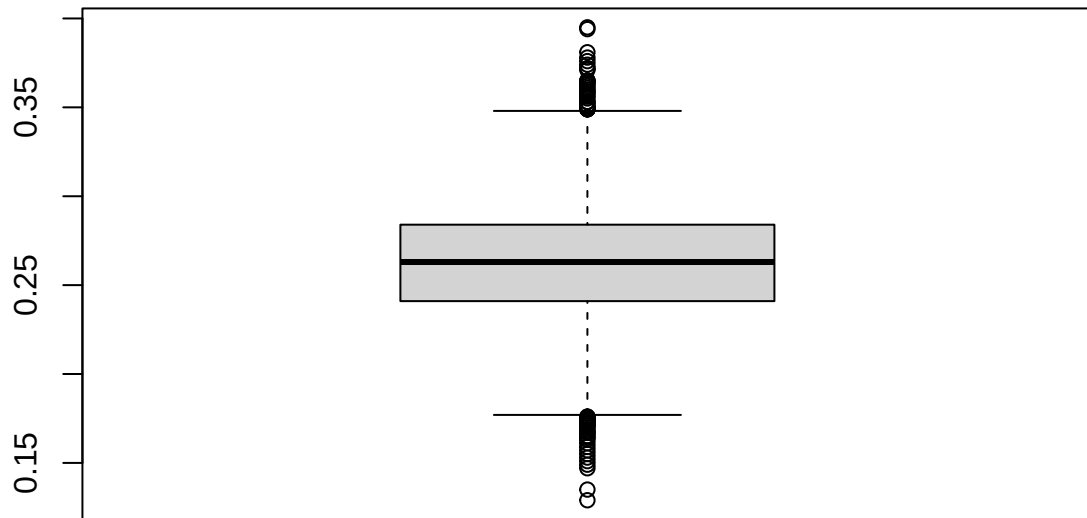
```
boxplot(Pitching_data$ERA)
```



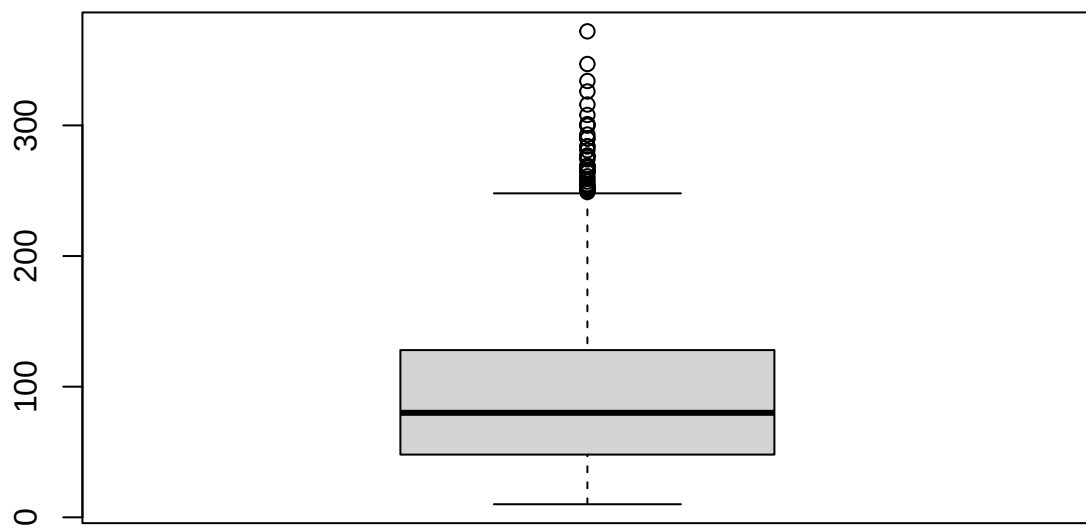
```
boxplot(Pitching_data$OutRate)
```



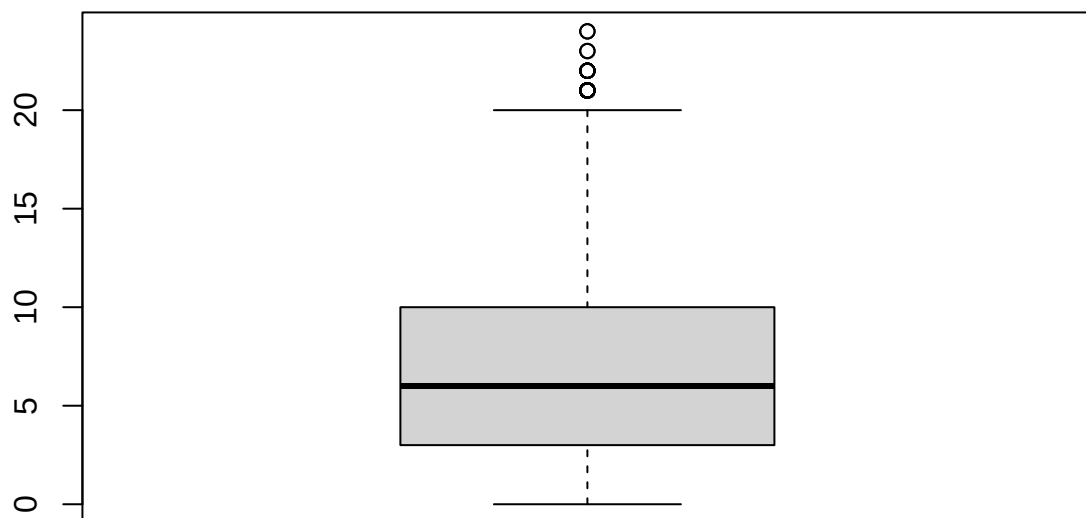
```
boxplot(Pitching_data$BAOpp)
```



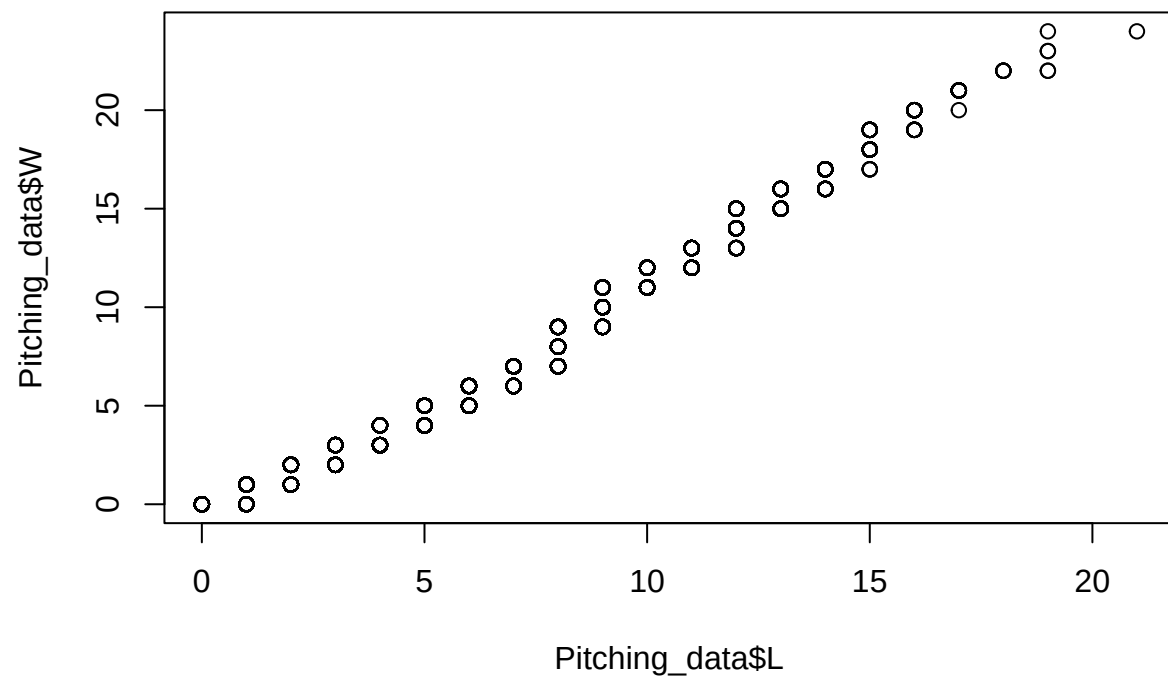
```
boxplot(Pitching_data$S0)
```



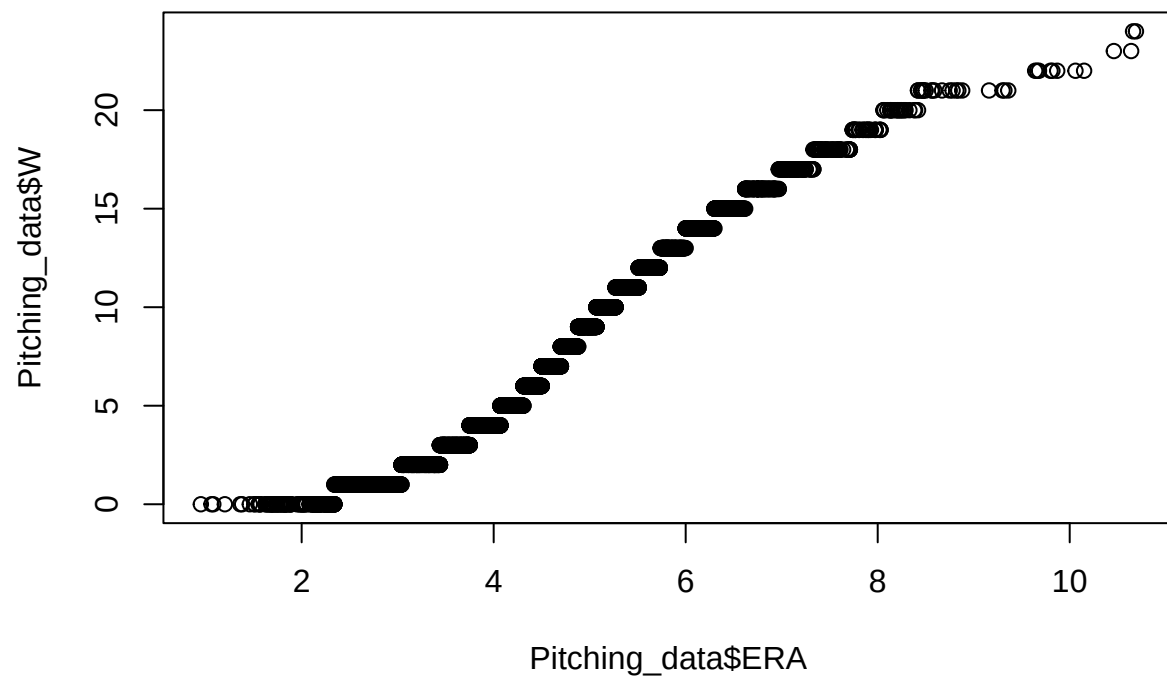
```
boxplot(Pitching_data$W)
```



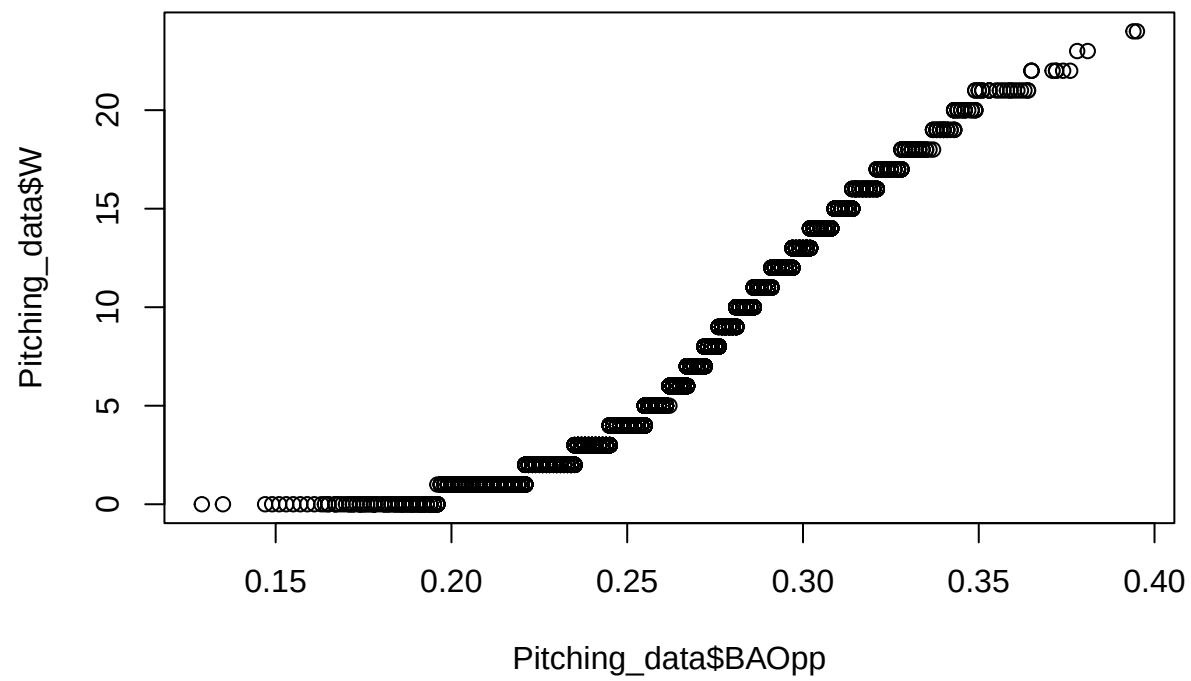
```
qqplot(Pitching_data$L, Pitching_data$W)
```



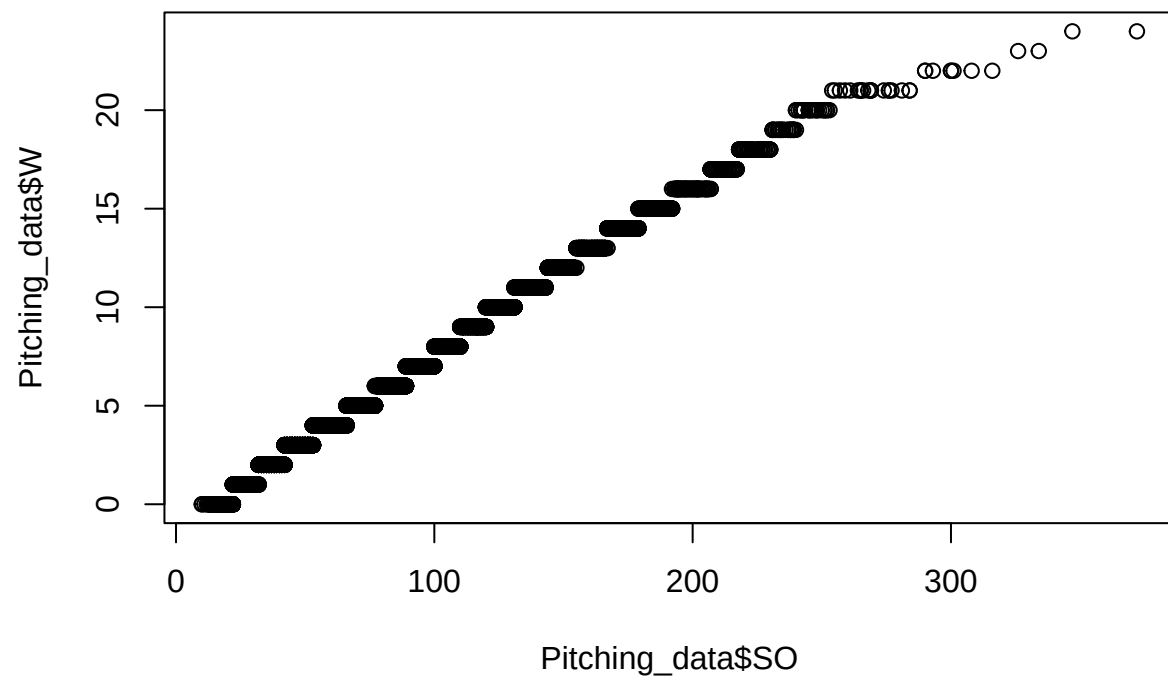
```
qqplot(Pitching_data$ERA, Pitching_data$W)
```

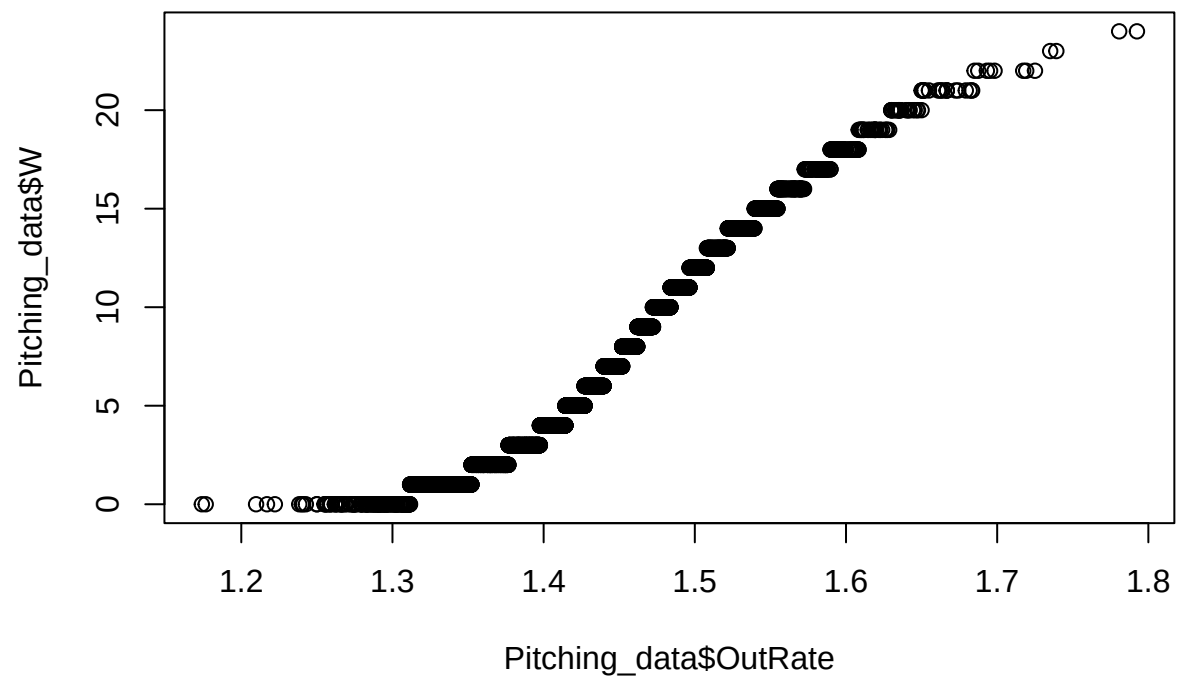
```
qqplot(Pitching_data$BAOpp, Pitching_data$W)
```



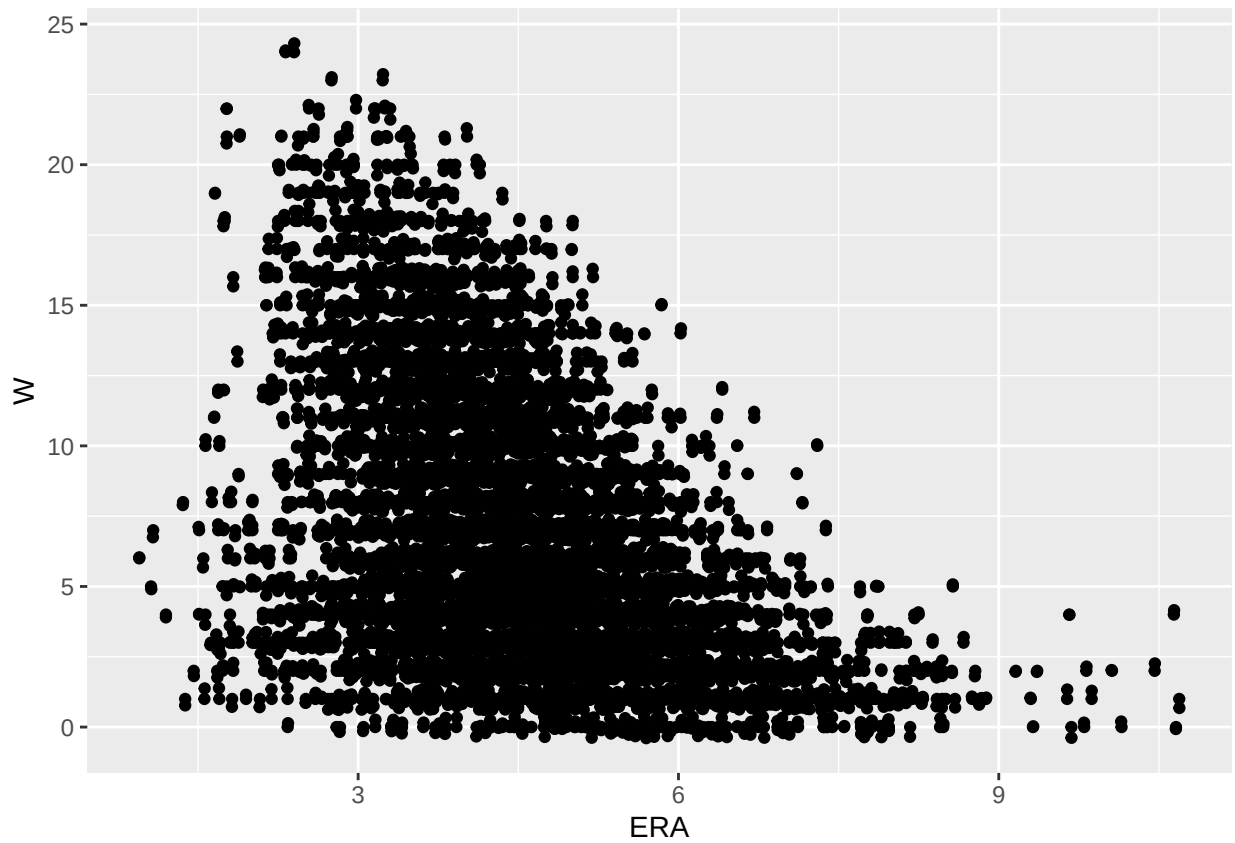
```
qqplot(Pitching_data$S0, Pitching_data$W)
```



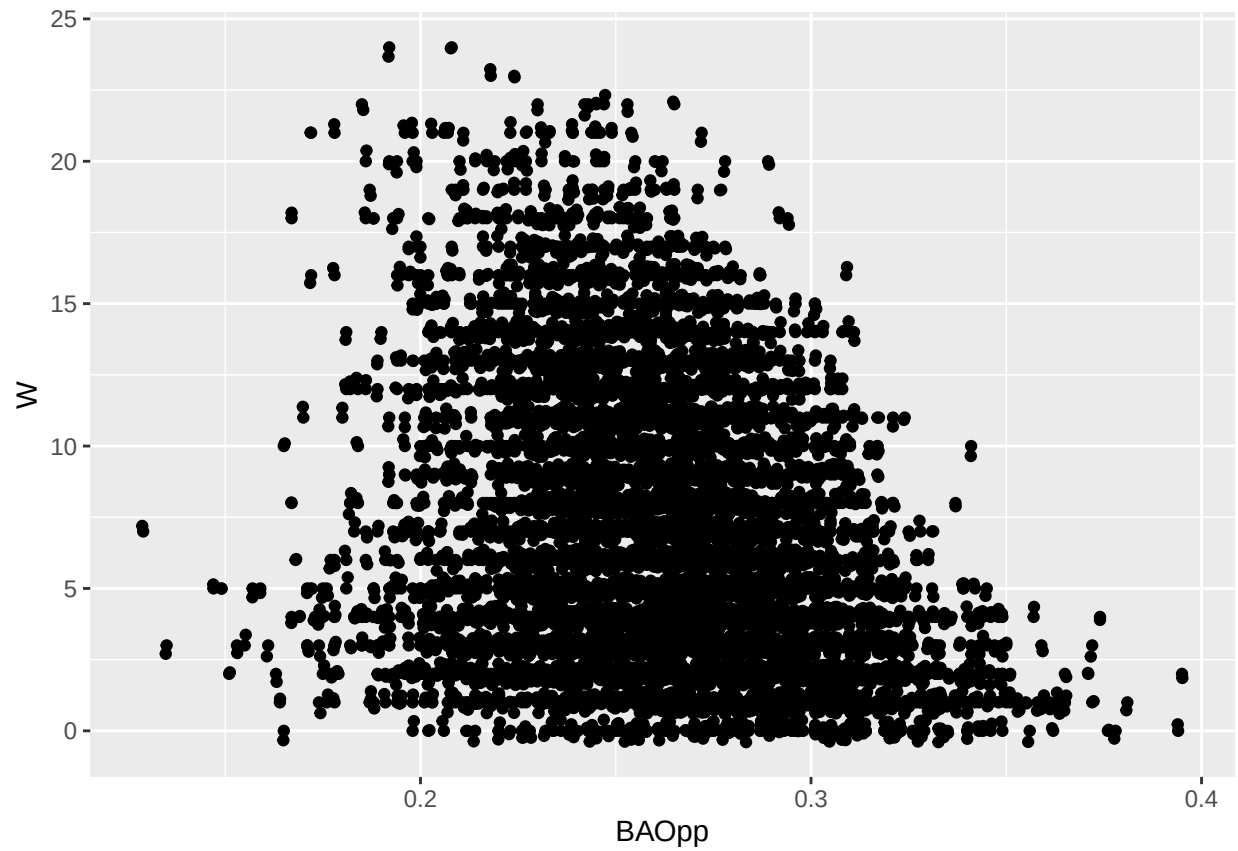
```
qqplot(Pitching_data$OutRate, Pitching_data$W)
```



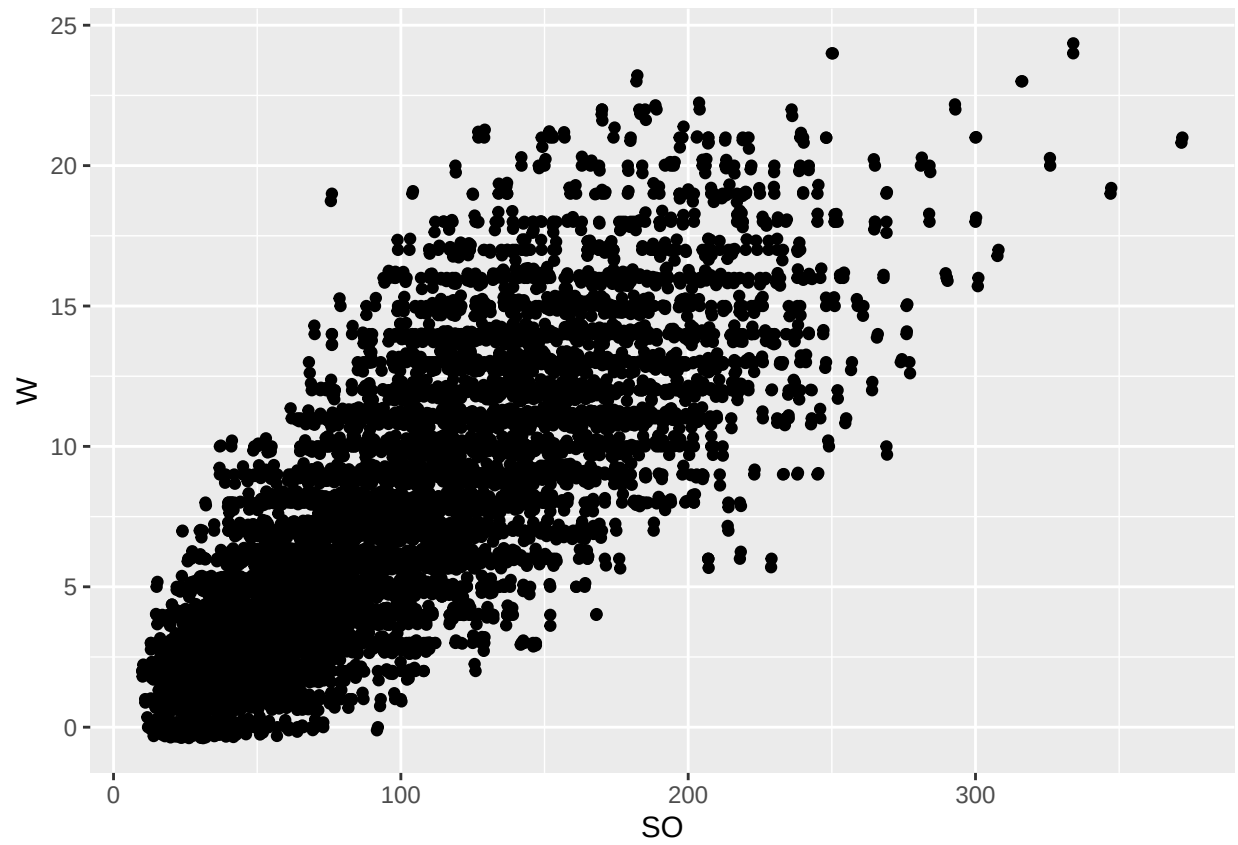
```
ggplot(Pitching_full, aes(x = ERA, y = W)) +  
  geom_point() +  
  geom_jitter()
```



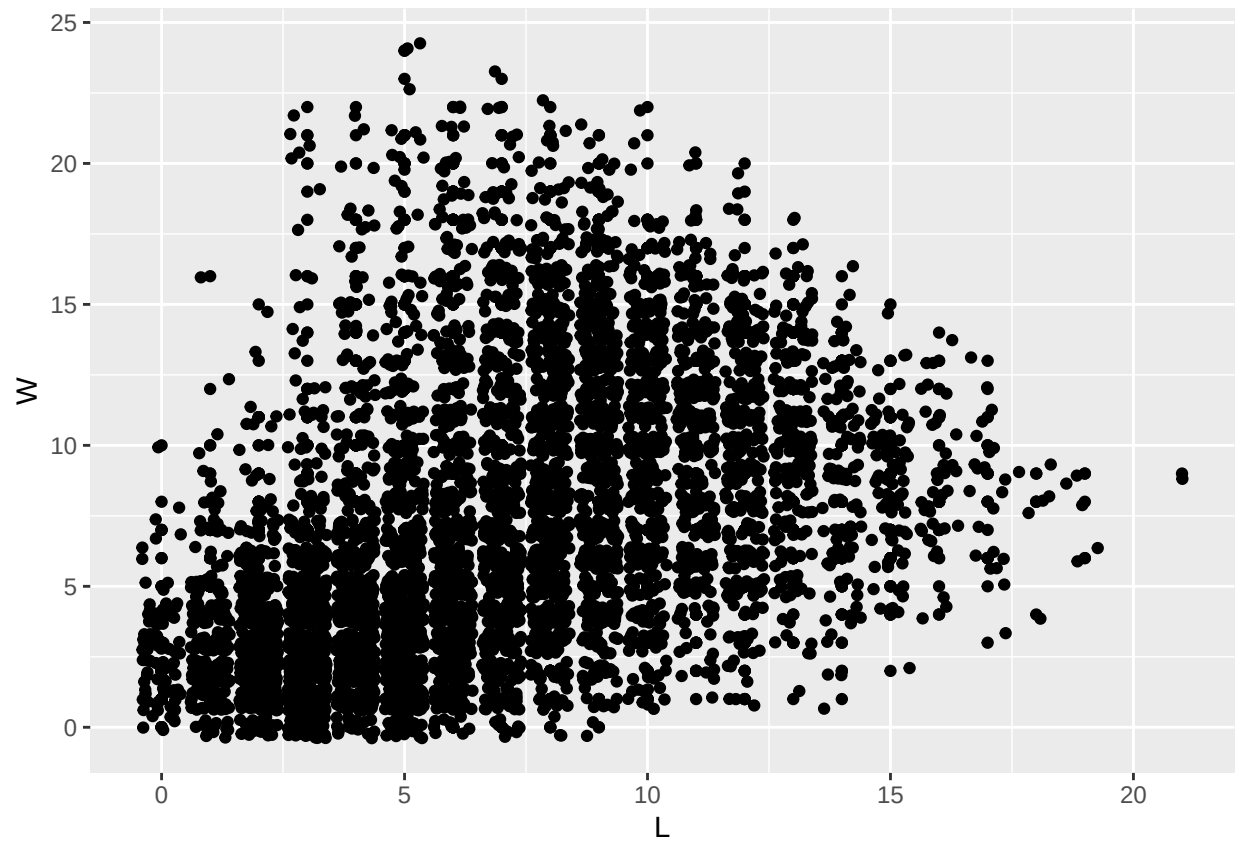
```
ggplot(Pitching_full, aes(x = BAOpp, y = W)) +  
  geom_point() +  
  geom_jitter()
```



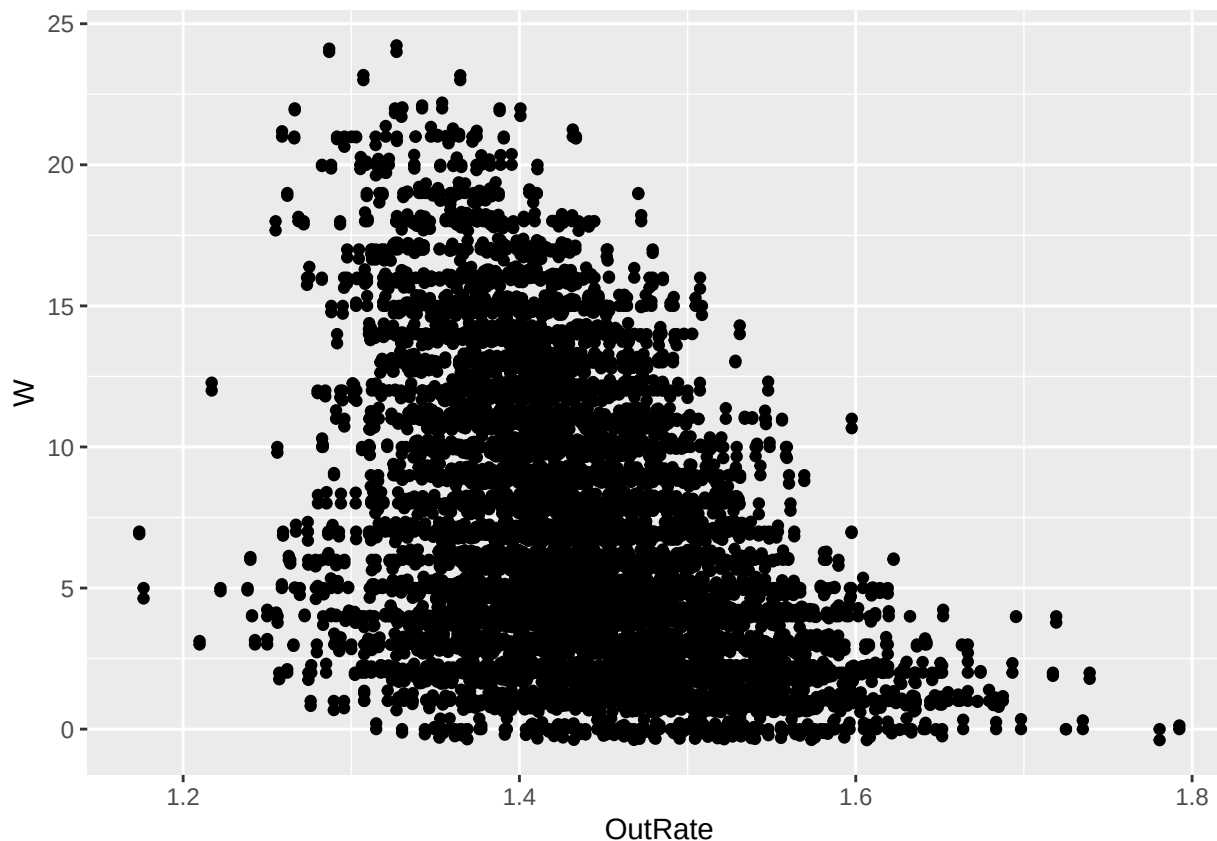
```
ggplot(Pitching_full, aes(x = S0, y = W)) +  
  geom_point() +  
  geom_jitter()
```



```
ggplot(Pitching_full, aes(x = L, y = W)) +  
  geom_point() +  
  geom_jitter()
```



```
ggplot(Pitching_full, aes(x = OutRate, y = W)) +  
  geom_point() +  
  geom_jitter()
```

Transformation Checks

```
summary(lm(Pitching_data$W ~ (Pitching_data$ERA))) #R^2 0.1955
```

```
##
## Call:
## lm(formula = Pitching_data$W ~ (Pitching_data$ERA))
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-11.1069	-3.1608	-0.5329	2.9451	14.0217

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	14.44068	0.20853	69.25	<2e-16 ***
Pitching_data\$ERA	-1.69114	0.04486	-37.70	<2e-16 ***

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.278 on 5843 degrees of freedom
## Multiple R-squared:  0.1956, Adjusted R-squared:  0.1955
## F-statistic: 1421 on 1 and 5843 DF, p-value: < 2.2e-16
```

```
summary(lm(Pitching_data$W ~ exp(Pitching_data$ERA))) #R^2 0.02186
```

```
##
## Call:
## lm(formula = Pitching_data$W ~ exp(Pitching_data$ERA))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.0007  -3.9276  -0.9788   3.0498  18.1771
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    7.0059250   0.0628691  111.44  <2e-16 ***
## exp(Pitching_data$ERA) -0.0005071  0.0000442  -11.47  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.717 on 5843 degrees of freedom
## Multiple R-squared:  0.02203, Adjusted R-squared:  0.02186
## F-statistic: 131.6 on 1 and 5843 DF, p-value: < 2.2e-16
```

```
summary(lm(Pitching_data$W ~ sqrt(Pitching_data$ERA))) #R^2 0.189
```

```
##
## Call:
## lm(formula = Pitching_data$W ~ sqrt(Pitching_data$ERA))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.4005  -3.1108  -0.5633   2.9463  14.0144
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    21.7322     0.4065   53.46  <2e-16 ***
## sqrt(Pitching_data$ERA) -7.0924     0.1921  -36.92  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.295 on 5843 degrees of freedom
## Multiple R-squared:  0.1892, Adjusted R-squared:  0.189
## F-statistic: 1363 on 1 and 5843 DF, p-value: < 2.2e-16
```

```
summary(lm(Pitching_data$W ~ log(Pitching_data$ERA))) #R^2 0.1756
```

```
##
## Call:
## lm(formula = Pitching_data$W ~ log(Pitching_data$ERA))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.8828  -3.0941  -0.6574   2.9870  14.1068
```

```
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      17.1513     0.2968   57.79  <2e-16 ***
## log(Pitching_data$ERA) -7.0432     0.1995  -35.30  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.331 on 5843 degrees of freedom
## Multiple R-squared:  0.1758, Adjusted R-squared:  0.1756
## F-statistic: 1246 on 1 and 5843 DF,  p-value: < 2.2e-16
```

```
summary(lm(Pitching_data$W ~ (Pitching_data$S0))) #R^2 0.669
```

```
##
## Call:
## lm(formula = Pitching_data$W ~ (Pitching_data$S0))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.5872  -1.8123  -0.2712   1.5209  13.3101
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      0.2768873  0.0704599     3.93 8.6e-05 ***
## Pitching_data$S0 0.0712242  0.0006553  108.70 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.744 on 5843 degrees of freedom
## Multiple R-squared:  0.6691, Adjusted R-squared:  0.669
## F-statistic: 1.181e+04 on 1 and 5843 DF,  p-value: < 2.2e-16
```

```
summary(lm(Pitching_data$W ~ exp(Pitching_data$S0))) #R^2 0.001332
```

```
##
## Call:
## lm(formula = Pitching_data$W ~ exp(Pitching_data$S0))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.8652  -3.8652  -0.8652   3.1348  17.1348
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      6.865e+00  6.235e-02 110.105 < 2e-16 ***
## exp(Pitching_data$S0) 3.915e-161 1.491e-161   2.626  0.00867 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.766 on 5843 degrees of freedom
## Multiple R-squared:  0.001503, Adjusted R-squared:  0.001332
## F-statistic: 8.792 on 1 and 5843 DF,  p-value: 0.003037
```

```
summary(lm(Pitching_data$W ~ sqrt(Pitching_data$S0))) #R^2 0.6716
```

```
##
## Call:
## lm(formula = Pitching_data$W ~ sqrt(Pitching_data$S0))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.1132 -1.8214 -0.1877  1.5780 12.8012
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -5.91580    0.12227  -48.38  <2e-16 ***
## sqrt(Pitching_data$S0)  1.38964    0.01271  109.33  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.733 on 5843 degrees of freedom
## Multiple R-squared:  0.6717, Adjusted R-squared:  0.6716
## F-statistic: 1.195e+04 on 1 and 5843 DF, p-value: < 2.2e-16
```

```
summary(lm(Pitching_data$W ~ log(Pitching_data$S0))) #R^2 0.6313
```

```
##
## Call:
## lm(formula = Pitching_data$W ~ log(Pitching_data$S0))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.9475 -1.9901 -0.2173  1.7588 12.1896
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -18.9660    0.2610  -72.66  <2e-16 ***
## log(Pitching_data$S0)  5.9520    0.0595  100.03  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.896 on 5843 degrees of freedom
## Multiple R-squared:  0.6313, Adjusted R-squared:  0.6313
## F-statistic: 1.001e+04 on 1 and 5843 DF, p-value: < 2.2e-16
```

```
summary(lm(Pitching_data$W ~ (Pitching_data$BAOpp))) #R^2 0.09017
```

```
##
## Call:
## lm(formula = Pitching_data$W ~ (Pitching_data$BAOpp))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.0837  -3.4996  -0.8588   3.1481  15.2344
```

```
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      18.2085     0.4746   38.37  <2e-16 ***
## Pitching_data$BAOpp -43.1810     1.7927  -24.09  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.55 on 5843 degrees of freedom
## Multiple R-squared:  0.09033, Adjusted R-squared:  0.09017
## F-statistic: 580.2 on 1 and 5843 DF, p-value: < 2.2e-16
```

```
summary(lm(Pitching_data$W ~ exp(Pitching_data$BAOpp))) #R^2 0.09244
```

```
##
## Call:
## lm(formula = Pitching_data$W ~ exp(Pitching_data$BAOpp))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.9552  -3.4833  -0.8404   3.1306  15.2117
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      50.575     1.791   28.24  <2e-16 ***
## exp(Pitching_data$BAOpp) -33.593     1.376  -24.42  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.544 on 5843 degrees of freedom
## Multiple R-squared:  0.0926, Adjusted R-squared:  0.09244
## F-statistic: 596.3 on 1 and 5843 DF, p-value: < 2.2e-16
```

```
summary(lm(Pitching_data$W ~ sqrt(Pitching_data$BAOpp))) #R^2 0.0852
```

```
##
## Call:
## lm(formula = Pitching_data$W ~ sqrt(Pitching_data$BAOpp))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.3587  -3.5148  -0.8711   3.1687  15.2750
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      28.6936     0.9366   30.64  <2e-16 ***
## sqrt(Pitching_data$BAOpp) -42.6756     1.8276  -23.35  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.562 on 5843 degrees of freedom
## Multiple R-squared:  0.08536, Adjusted R-squared:  0.0852
## F-statistic: 545.3 on 1 and 5843 DF, p-value: < 2.2e-16
```

```
summary(lm(Pitching_data$W ~ log(Pitching_data$BAOpp))) #R^2 0.07971
```

```
##
## Call:
## lm(formula = Pitching_data$W ~ log(Pitching_data$BAOpp))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.6242  -3.5062  -0.8943   3.1783  15.3112
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      -7.1453     0.6251  -11.43  <2e-16 ***
## log(Pitching_data$BAOpp) -10.4170     0.4625  -22.52  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.576 on 5843 degrees of freedom
## Multiple R-squared:  0.07987, Adjusted R-squared:  0.07971
## F-statistic: 507.2 on 1 and 5843 DF, p-value: < 2.2e-16
```

```
summary(lm(Pitching_data$W ~ (Pitching_data$OutRate))) #R^2 0.192
```

```
##
## Call:
## lm(formula = Pitching_data$W ~ (Pitching_data$OutRate))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.6147  -3.1478  -0.5547   2.9565  14.0710
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      49.3083     1.1399   43.26  <2e-16 ***
## Pitching_data$OutRate -29.5437     0.7925  -37.28  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.287 on 5843 degrees of freedom
## Multiple R-squared:  0.1921, Adjusted R-squared:  0.192
## F-statistic: 1390 on 1 and 5843 DF, p-value: < 2.2e-16
```

```
summary(lm(Pitching_data$W ~ exp(Pitching_data$OutRate))) #R^2 0.1927
```

```
##
## Call:
## lm(formula = Pitching_data$W ~ exp(Pitching_data$OutRate))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.2676  -3.1838  -0.5563   2.9431  14.0503
```

```
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      35.9654     0.7807   46.07  <2e-16 ***
## exp(Pitching_data$OutRate) -6.9005     0.1847  -37.37  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.285 on 5843 degrees of freedom
## Multiple R-squared:  0.1929, Adjusted R-squared:  0.1927
## F-statistic: 1396 on 1 and 5843 DF, p-value: < 2.2e-16
```

```
summary(lm(Pitching_data$W ~ sqrt(Pitching_data$OutRate))) #R^2 0.1913
```

```
##
## Call:
## lm(formula = Pitching_data$W ~ sqrt(Pitching_data$OutRate))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.8443  -3.1486  -0.5626   2.9589  14.0875
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)       91.914     2.287   40.19  <2e-16 ***
## sqrt(Pitching_data$OutRate) -70.979     1.908  -37.19  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.289 on 5843 degrees of freedom
## Multiple R-squared:  0.1914, Adjusted R-squared:  0.1913
## F-statistic: 1383 on 1 and 5843 DF, p-value: < 2.2e-16
```

```
summary(lm(Pitching_data$W ~ log(Pitching_data$OutRate))) #R^2 0.1904
```

```
##
## Call:
## lm(formula = Pitching_data$W ~ log(Pitching_data$OutRate))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.129  -3.143  -0.569   2.950  14.105
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)       22.2358     0.4182   53.17  <2e-16 ***
## log(Pitching_data$OutRate) -42.5668     1.1477  -37.09  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.292 on 5843 degrees of freedom
## Multiple R-squared:  0.1906, Adjusted R-squared:  0.1904
## F-statistic: 1375 on 1 and 5843 DF, p-value: < 2.2e-16
```

```
summary(lm(Pitching_data$W ~ (Pitching_data$L))) #R^2 0.2236
```

```
##
## Call:
## lm(formula = Pitching_data$W ~ (Pitching_data$L))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.3033  -2.9821  -0.8124   2.1876  18.1876
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.76194    0.11415   24.20  <2e-16 ***
## Pitching_data$L 0.61010    0.01487   41.04  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.203 on 5843 degrees of freedom
## Multiple R-squared:  0.2238, Adjusted R-squared:  0.2236
## F-statistic: 1684 on 1 and 5843 DF,  p-value: < 2.2e-16
```

```
summary(lm(Pitching_data$W ~ exp(Pitching_data$L))) #R^2 -1.771e-05
```

```
##
## Call:
## lm(formula = Pitching_data$W ~ exp(Pitching_data$L))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.8654  -3.8654  -0.8654   3.1346  17.1346
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    6.865e+00  6.243e-02 109.966  <2e-16 ***
## exp(Pitching_data$L) 3.244e-09  3.426e-09   0.947    0.344
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.77 on 5843 degrees of freedom
## Multiple R-squared:  0.0001534, Adjusted R-squared: -1.771e-05
## F-statistic: 0.8965 on 1 and 5843 DF,  p-value: 0.3438
```

```
summary(lm(Pitching_data$W ~ sqrt(Pitching_data$L))) #R^2 0.2306
```

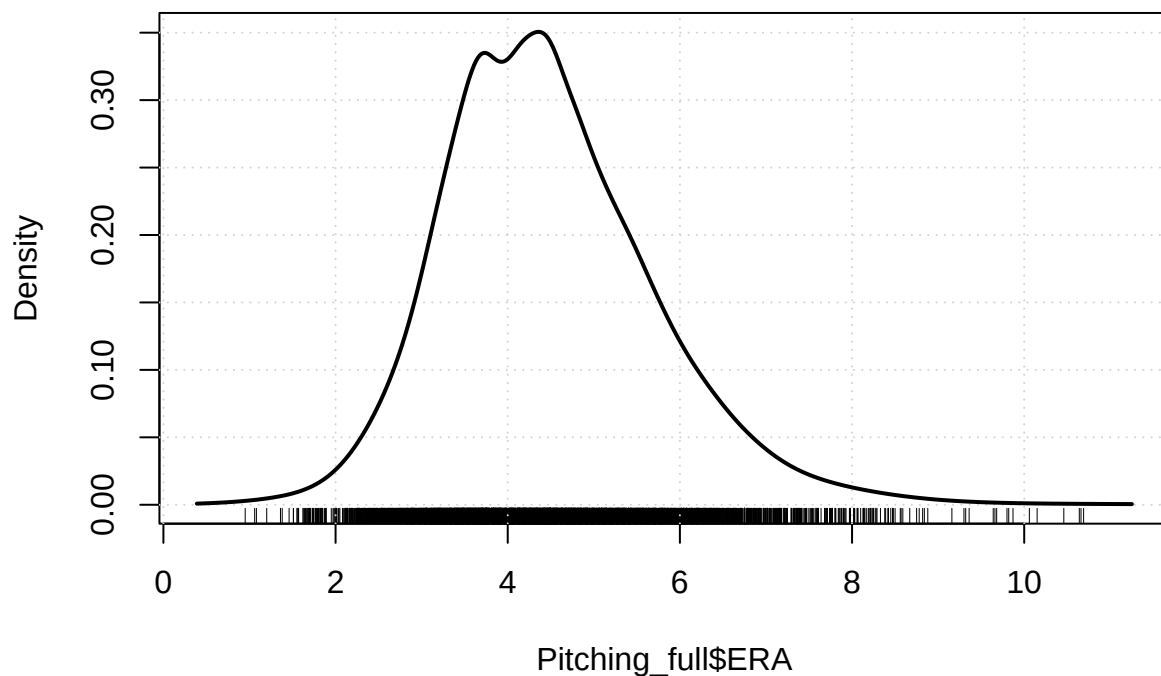
```
##
## Call:
## lm(formula = Pitching_data$W ~ sqrt(Pitching_data$L))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.5751  -2.9066  -0.7641   2.2970  17.8243
```



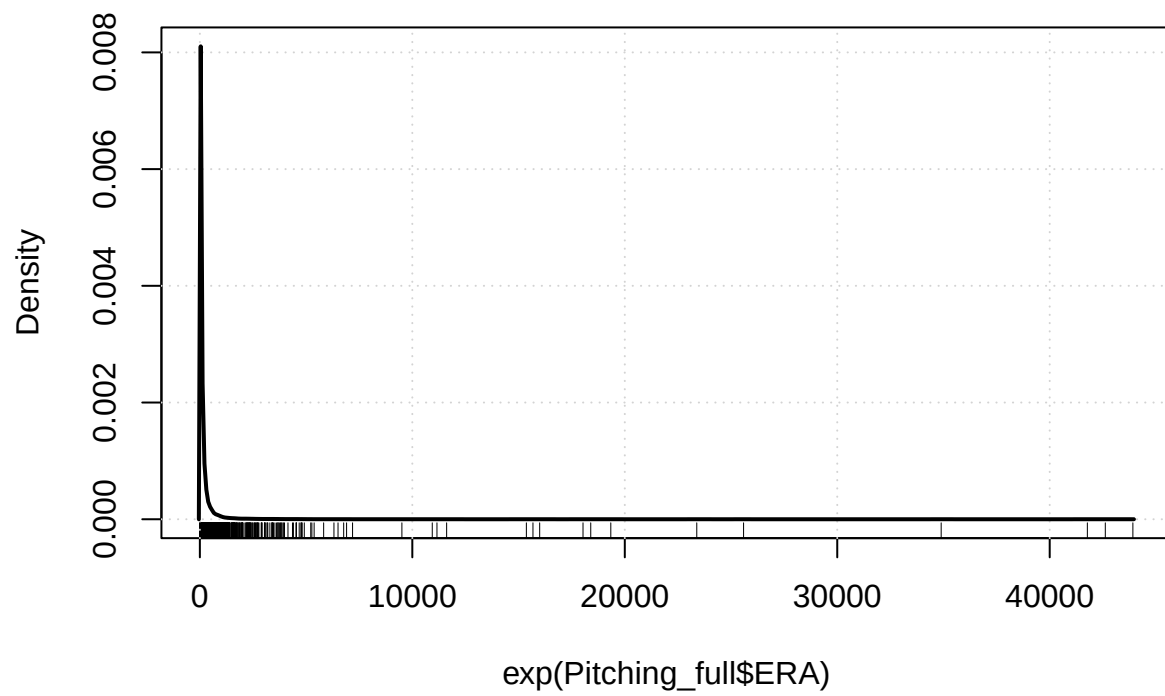
```
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -0.35812    0.18109  -1.978   0.048 *
## sqrt(Pitching_data$L)  2.92202    0.06981  41.858  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.184 on 5843 degrees of freedom
## Multiple R-squared:  0.2307, Adjusted R-squared:  0.2306
## F-statistic: 1752 on 1 and 5843 DF, p-value: < 2.2e-16
```

```
#summary(lm(Pitching_data$W ~ log(Pitching_data$L))) No R^2 because log(0)
```

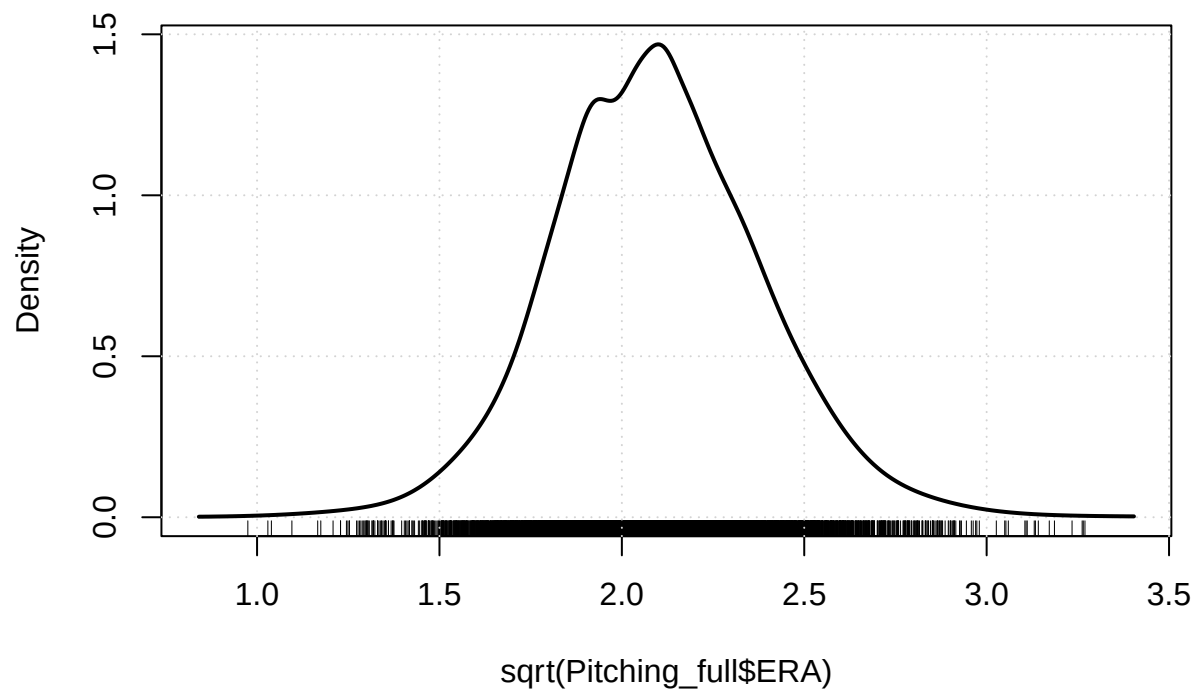
```
densityPlot(Pitching_full$ERA) #Best Visually and Best R^2
```



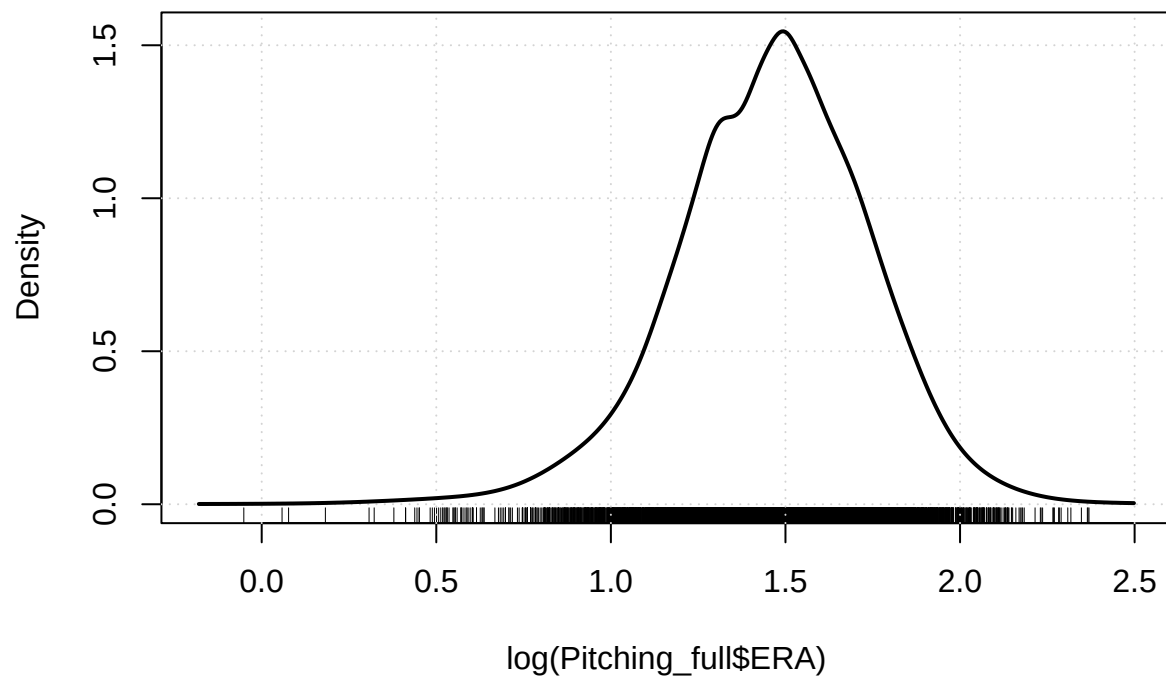
```
densityPlot(exp(Pitching_full$ERA))
```



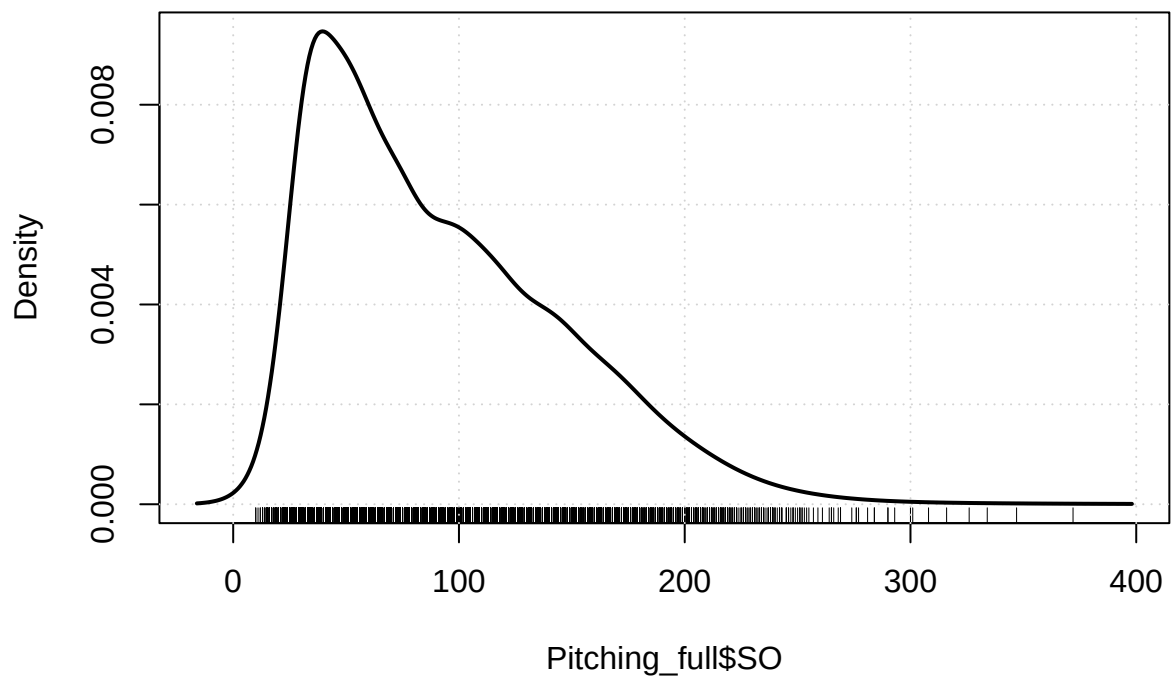
```
densityPlot(sqrt(Pitching_full$ERA))
```



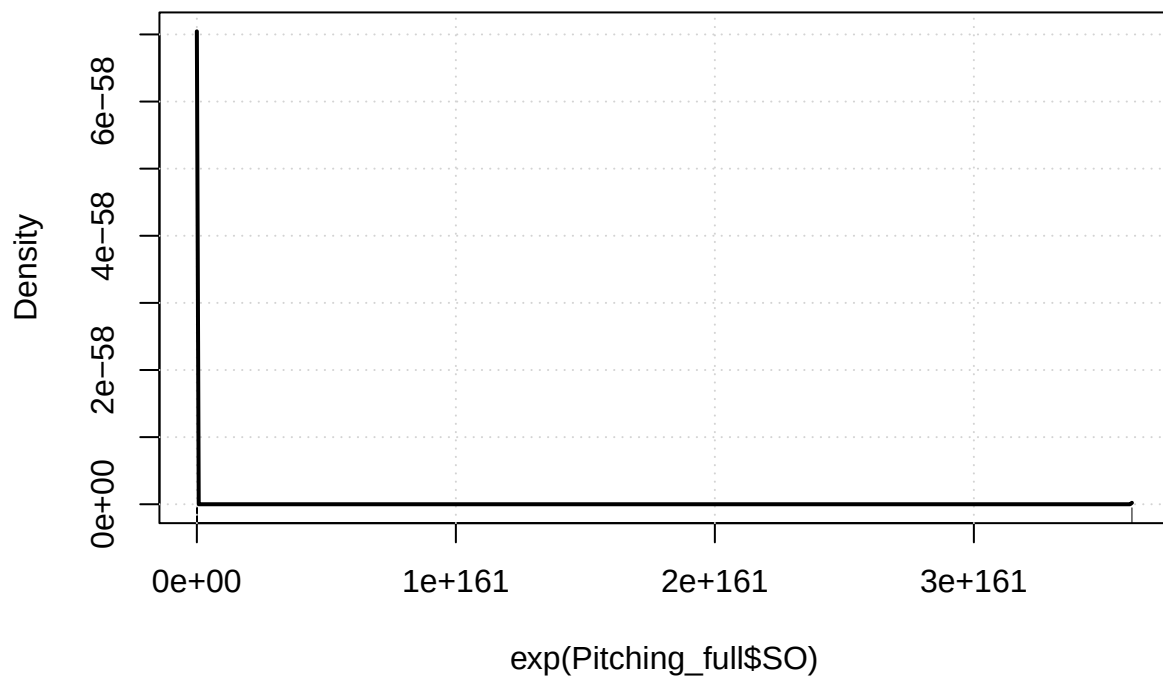
```
densityPlot(log(Pitching_full$ERA))
```



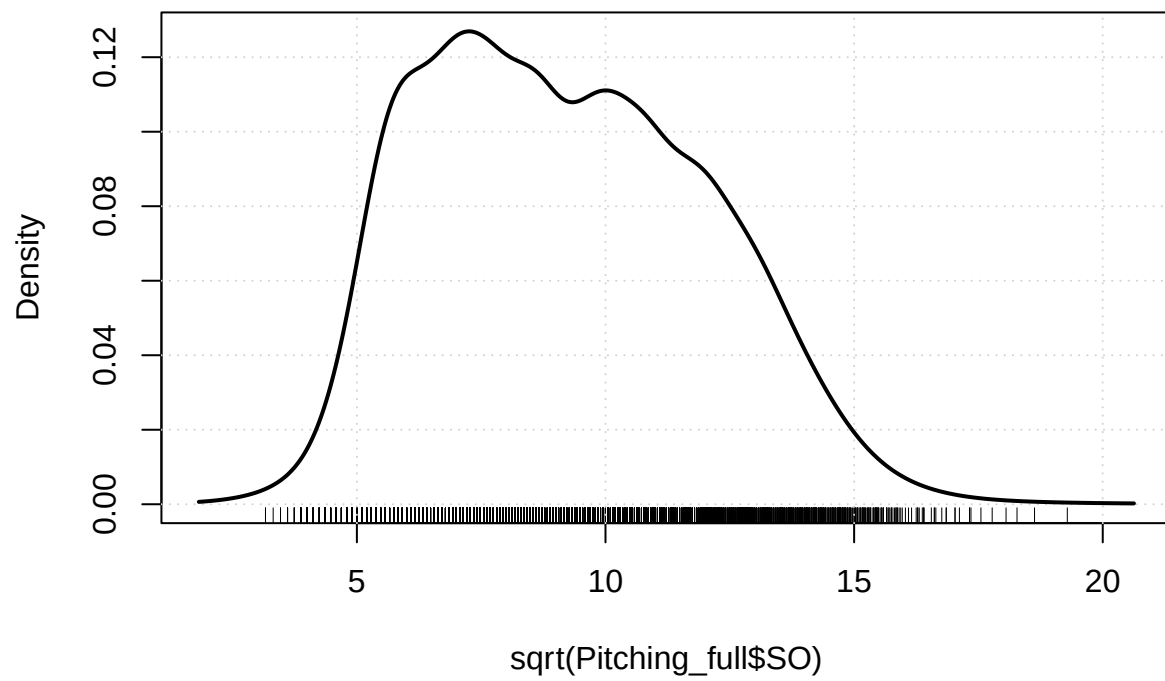
```
densityPlot(Pitching_full$S0)
```



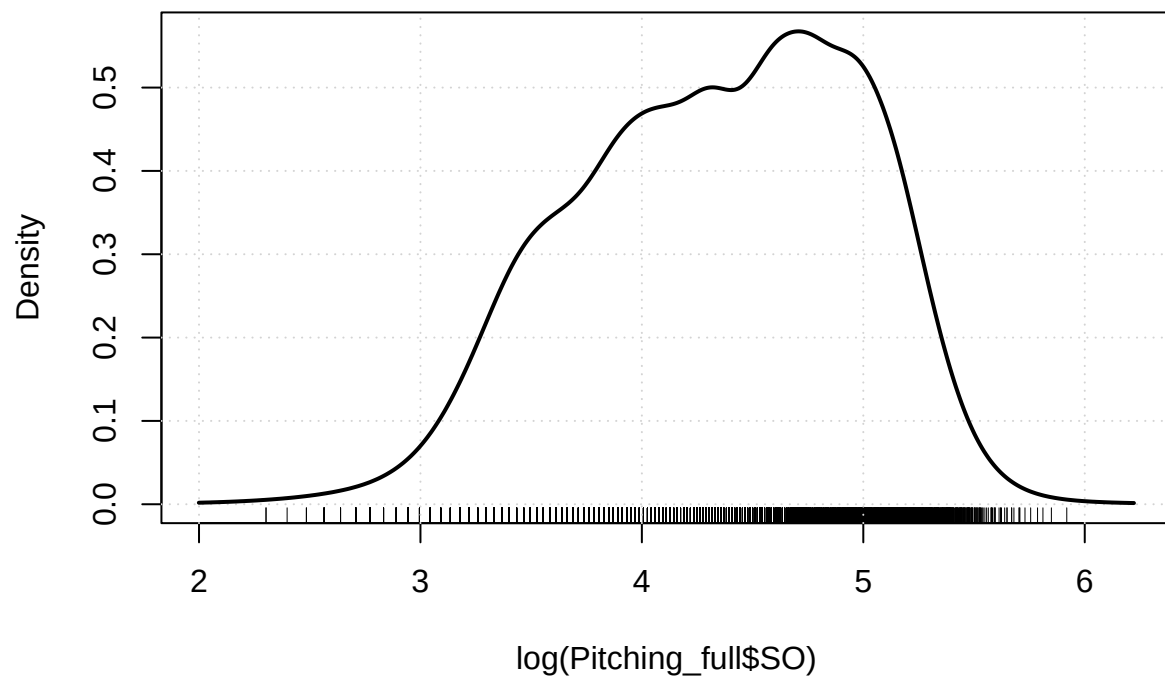
```
densityPlot(exp(Pitching_full$SO))
```



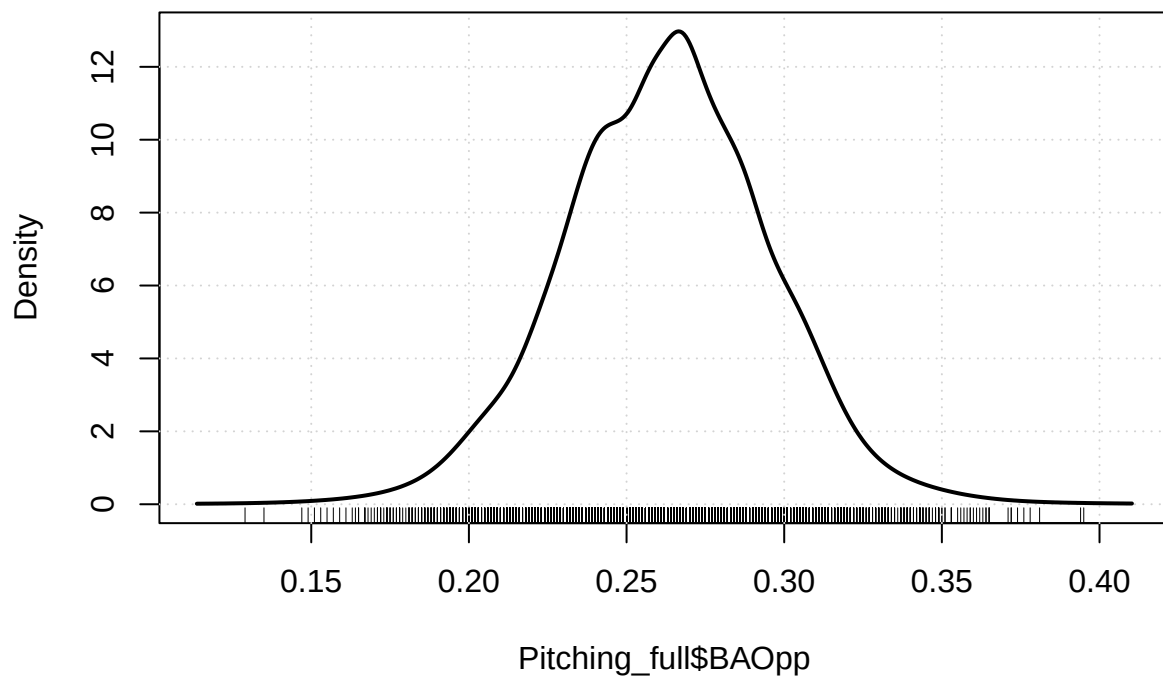
```
densityPlot(sqrt(Pitching_full$SO)) #Best Visually, Best R^2
```



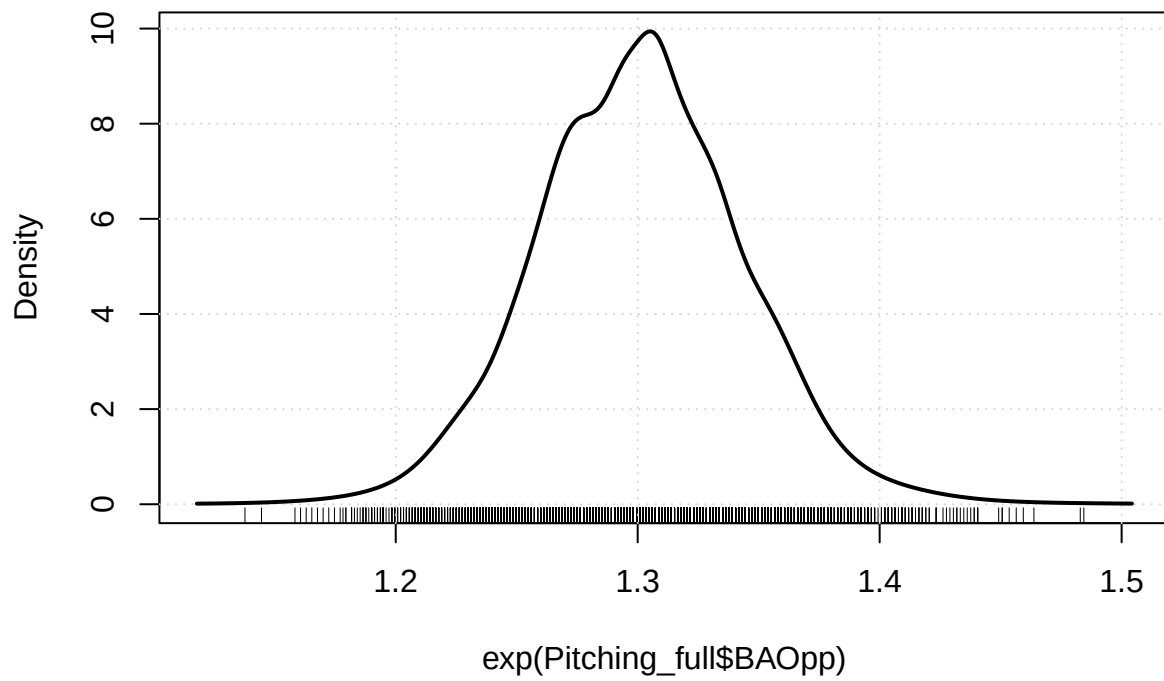
```
densityPlot(log(Pitching_full$SO))
```



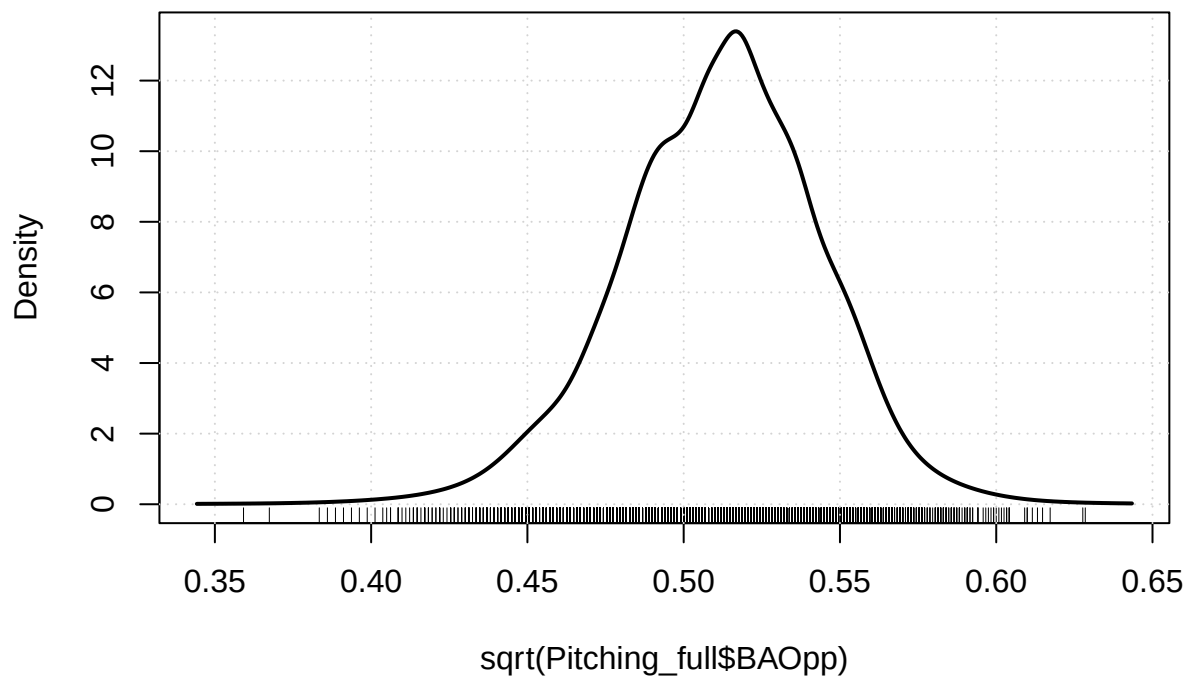
```
densityPlot(Pitching_full$BAOpp)
```

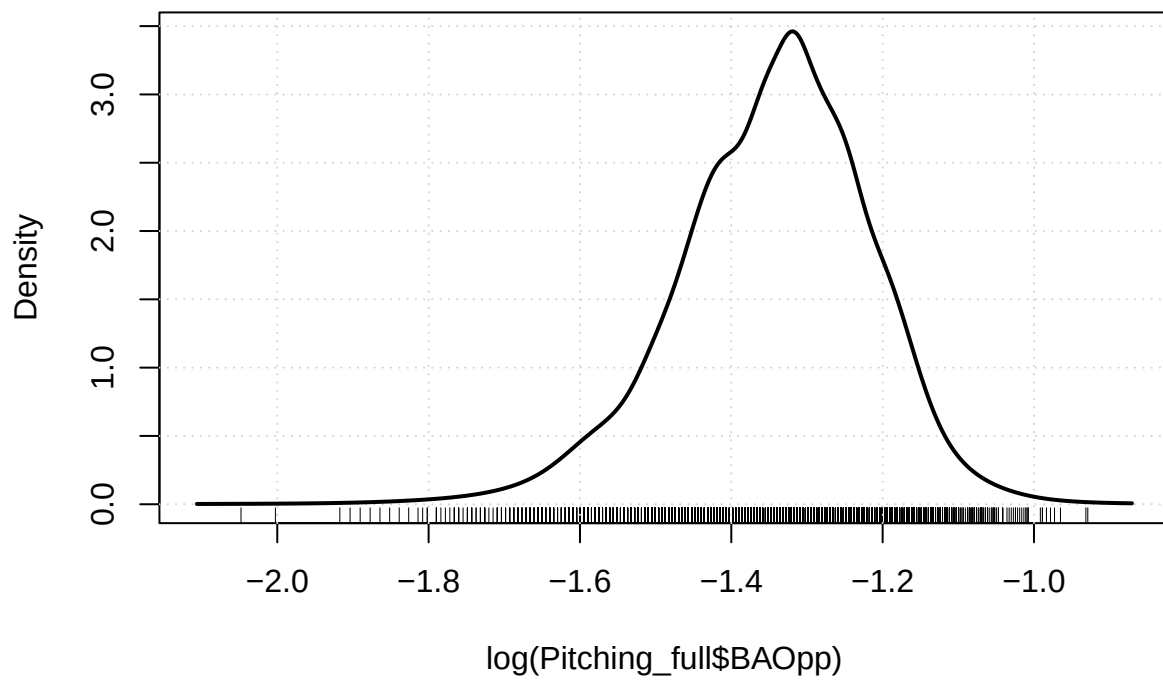
```
densityPlot(exp(Pitching_full$BAOpp)) #Best Visually, Best R^2
```



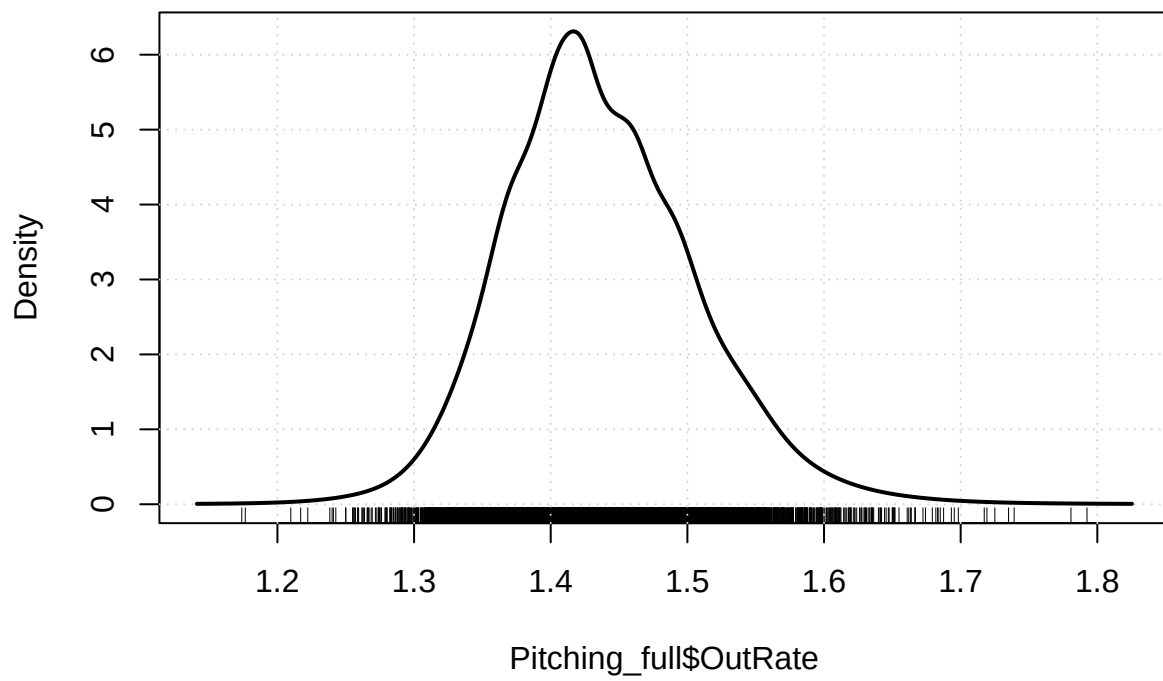
```
densityPlot(sqrt(Pitching_full$BAOpp))
```



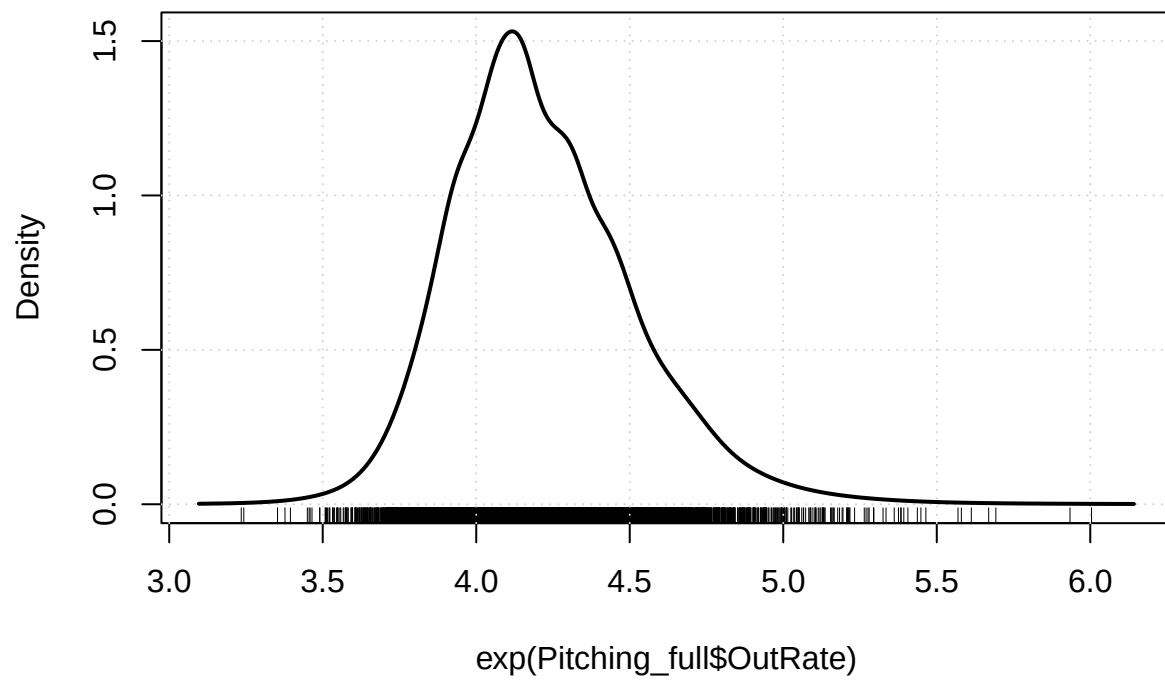
```
densityPlot(log(Pitching_full$BAOpp))
```



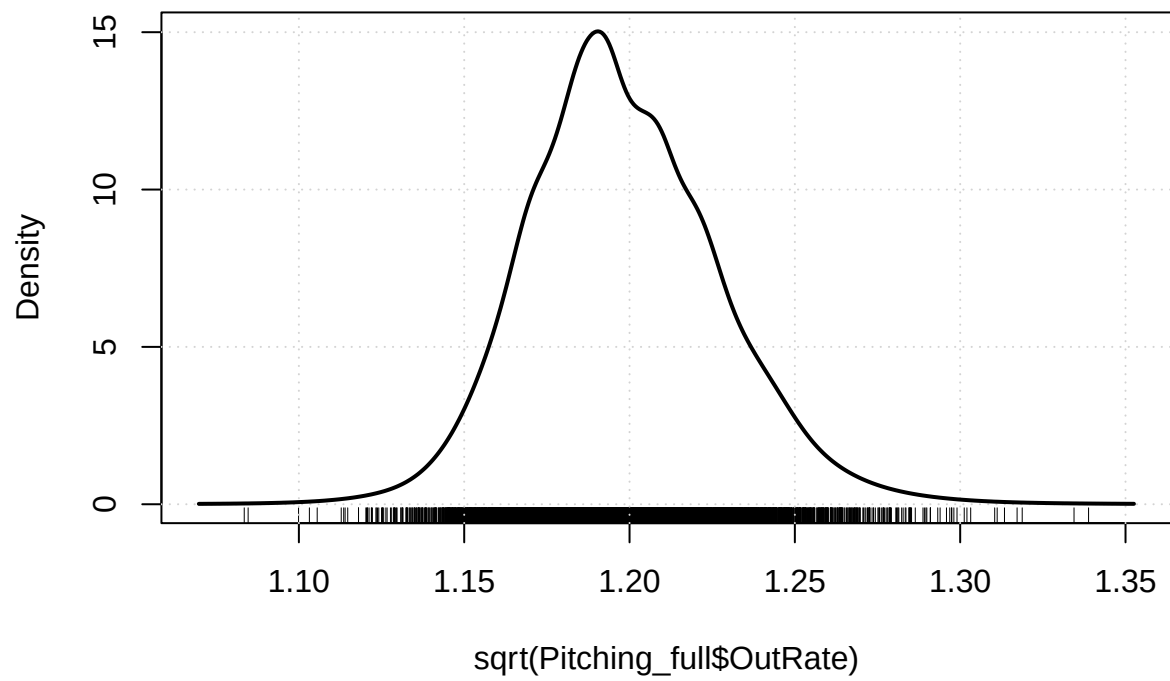
```
densityPlot(Pitching_full$OutRate)
```



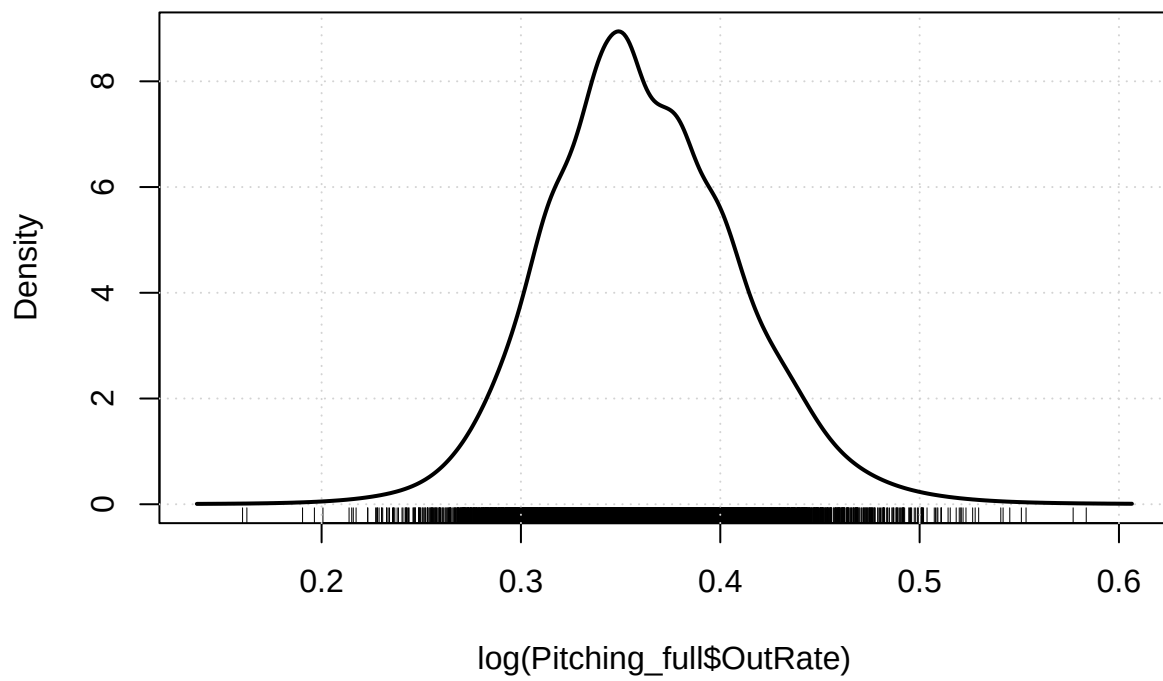
```
densityPlot(exp(Pitching_full$OutRate))
```



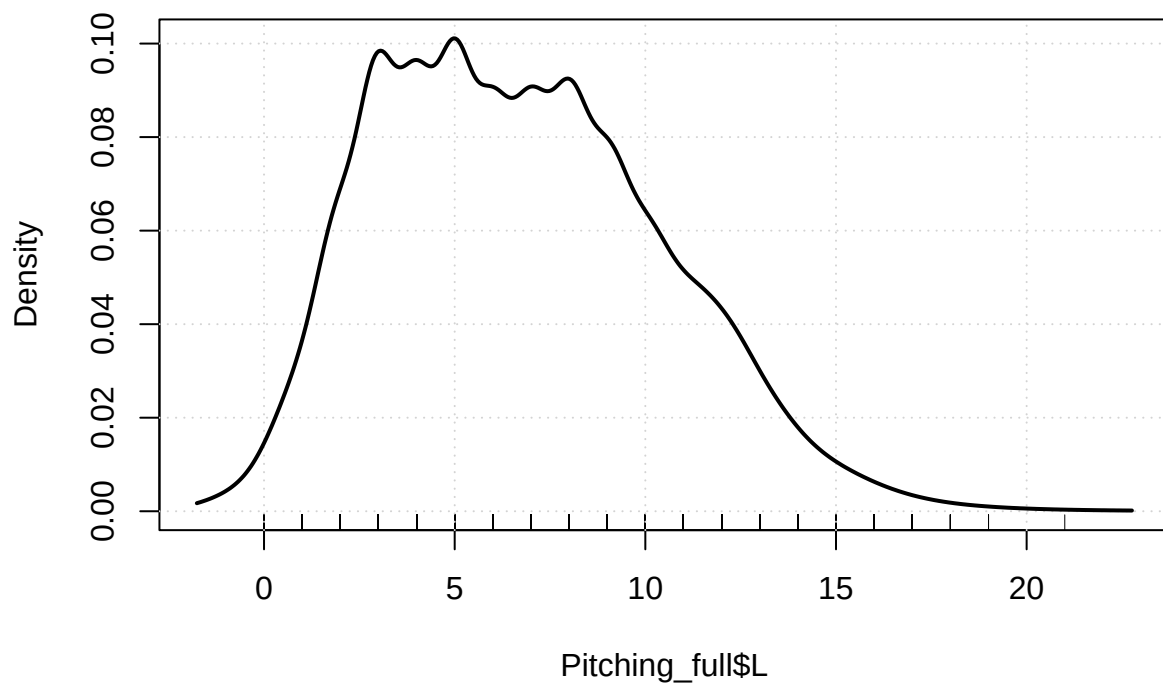
```
densityPlot(sqrt(Pitching_full$OutRate)) #Best Visually, Comparable R2
```



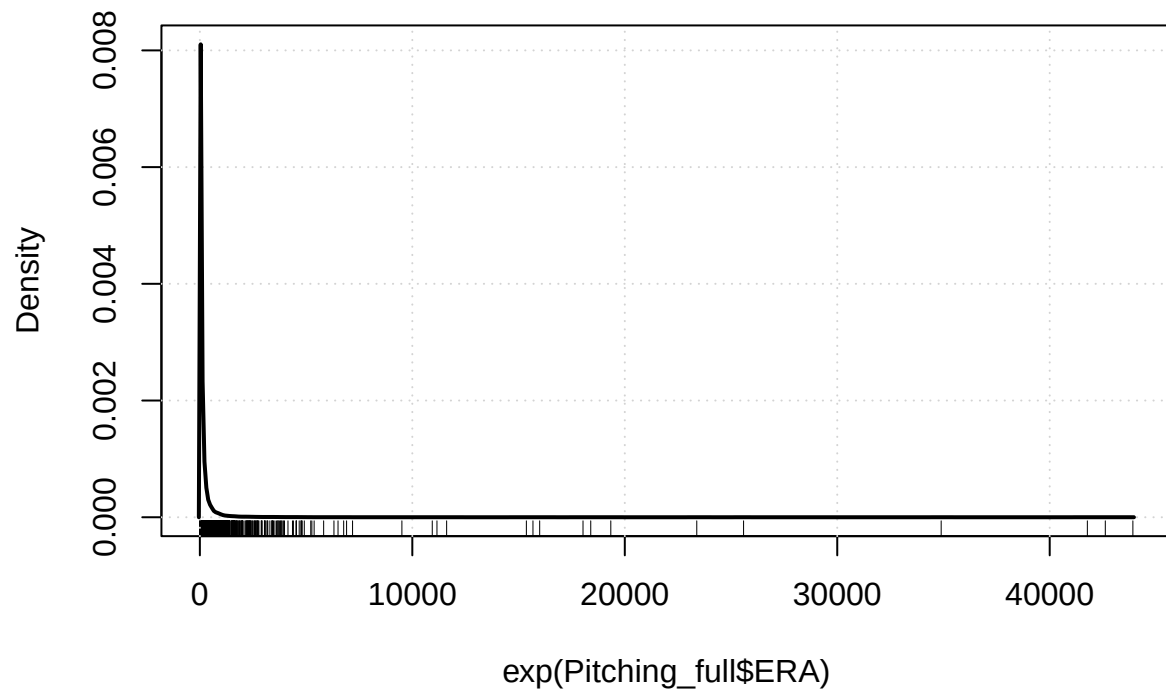
```
densityPlot(log(Pitching_full$OutRate))
```



```
densityPlot(Pitching_full$L) #Best Visually, Comparable R2 (Use This One)
```

```
densityPlot(exp(Pitching_full$ERA))
```



```
densityPlot(sqrt(Pitching_full$L)) #Best  $R^2$ 
```

